MASTERS OF BUSINESS ADMINISTRATION

SEMESTER 1



DMBA115

MANAGERIAL ECONOMICS

Unit - 4

Production Analysis & Cost Analysis

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1. INTRODUCTION

In the previous unit, we explored how demand and supply interact to achieve market equilibrium. We also learned that production involves converting raw materials or semi-finished inputs into goods or services. In this unit, we will delve into the production process and its associated challenges.

A business firm is both an economic and a production unit, with production being central to its operations. The main goal of production is to meet the demand for goods and services in the market. However, output doesn't always increase in direct proportion to the increase in inputs. We also learned about tools to determine the optimal combinations of inputs needed to produce desired outputs, along with the advantages and disadvantages of large-scale production.

Since production uses scarce inputs, costs are incurred during the process. As businesses aim for profit maximization, controlling costs becomes essential. In this unit, we will focus on cost analysis and how firms can manage their costs effectively. Costs are analyzed from the producer's perspective and are measured in monetary terms to aid management decisions.

In the production process, producers compensate for the services provided by factor inputs, and these compensations determine the cost of production. The total expenses, both explicit and implicit, incurred in transforming inputs into outputs are considered the cost of production. Understanding cost concepts and the cost-output relationship is key in cost analysis.

1.1 Objectives



After studying this unit, you should be able to:

- Explain the concept of production, production function and its managerial applications.
- Analyse short run and long run production function with illustrations.
- Describe the various dimensions, advantages and demerits of large-scale production.
- Define concepts of production costs and assess their managerial applications
- Apply the short run and long run cost-output relationships in arriving at the production decision

2. PRODUCTION AND PRODUCTION FUNCTION

In this section, we will discuss the concept of production and the production function. Figure 5.1 depicts the concept of production.

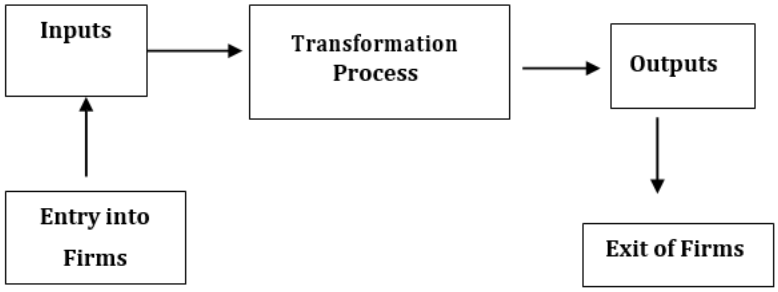


Figure 1: Concept of Production

The term "production" means transformation of physical "inputs" into physical "outputs".

The term "inputs" or "factors of production" includes all items or those things which are required by the firm to produce a particular product. There are four general factors in production like land, labor, capital and Entrepreneur. Along with those four factors there are some more factors added to the factors of production, for example inputs also include other items like raw materials of all kinds, power, fuel, water, technology, time and services like transport and communications, warehousing, marketing, banking, shipping and insurance. It also includes the ability, talents, capacities, knowledge, experience and wisdom of human beings. Thus, the term "inputs" has a wider scope in economics. What we get at the end of production process is called "outputs". In short, "outputs" refer to finished products.

Production always results in either the creation of new utilities or the enhancement of value. It is an activity aimed at increasing the consumer satisfaction derived from goods and services. Producers undertake production by incurring various costs. Production analysis is then conducted in physical terms to establish the relationship between the physical inputs used and the physical outputs generated.

Higher levels of production are an index of an organization's and society's progress and growth. They lead to higher income, employment, and economic prosperity. The production of different types of goods and services in different nations indicates the nature of economic interdependence between them.

Production function

The entire theory of production centres revolves around the concept of the production function. A "production function" expresses the relationship between the physical quantity of inputs employed and the physical quantity of outputs obtained by a firm. It specifies a flow of output resulting from a flow of inputs during a specified period of time. It may be in the form of a table, a graph or an equation specifying the maximum output rate from a given amount of inputs used. As it relates inputs to outputs, it is also called "input-output relation." The production is purely physical in nature and is determined by the quantum of technology, availability of equipment, labor, raw materials etc. employed by a firm.

A production function can be represented in the form of a mathematical model or equation as Q = f (L, N, K…. etc.) where Q stands for quantity of output per unit of time and L, N, K etc. are the various factor inputs like land, capital, labor, etc. which are used in the production of output. The rate of output Q is thus, a function of the factor inputs L, N, K etc., employed by the firm per unit of time.

Factor inputs are of two types as follows:

1. ***Fixed inputs*** **–** Fixed inputs are those factors the quantity of which remains constant irrespective of the level of output produced by a firm, for example, land, buildings, machines, tools, equipment, superior types of labor, top management, etc.
2. ***Variable inputs*** **–** Variable inputs are those factors, the quantity of which varies with variations in the levels of output produced by a firm, for example, raw materials, power, fuel, water, transport, communication, etc.

The distinction between the two will hold good only in the short run. In the long run, all factor inputs will become variable in nature.

The short run is a period of time in which the amount of some inputs called fixed factors cannot be changed. Here only the variable factors can be varied while fixed factors like plants, machinery, top management etc. would remain constant. Time available at the disposal of a producer to make changes in the quantum of fixed factor inputs is limited in the short run.

Long run is a period of time in which the quantities of all factors of production can be varied and there are no fixed factors. Here the producer will have adequate time to make changes in the fixed and variable factor combinations. It is necessary to note that the production function is assumed to be a continuous function, i.e., it is assumed that a change in any of the variable factors, produces

corresponding changes in the output.

Generally speaking, there are two types of production functions. They are as follows:

1. Short run production function – In this case, the producer will keep all fixed factors as constant and change only a few variable factor inputs. In the short run, we come across two kinds of production functions:
    - Quantities of all inputs both fixed and variable will be kept constant and only one variable input will be varied, for example, law of variable proportions.
    - Quantities of all factor inputs are kept constant and only two variable factor inputs are varied, for example, iso-quants and iso-cost curves.
2. Long run production function – In this case, the producer will vary the quantities of all factor inputs, both fixed as well as variable in the same proportion, for example, the laws of returns to scale.

Each firm has its own production function which is determined by the state of technology, managerial ability, organizational skills, etc. of a firm. It may be in the following manner:

1. The quantity of inputs may be reduced while the quantity of output may remain the same.
2. The quantity of input may be reduced while the quantity of output may increase.
3. The quantity of inputs may be kept constant while the quantity of output may increase.

If there are any improvements in the firm, the old production function is disturbed and a new one takes its place.

Uses of production function

While the production function may seem abstract and theoretical, it is both logical and practical in real-world applications. It provides significant value to managers and executives, serving as a vital tool in the decision-making process at the firm level.

There are several possible combinations of inputs and decision makers have to choose the most appropriate among them. The following are some of the important uses of production function:

1. It can be used to calculate or work out the least cost input combination for a given output or the maximum output-input combination for a given cost.
2. It is useful in working out an optimal and economic combination of inputs for getting a certain level of output. The utility of employing a unit of variable factor input in the production process can be better judged with the help of production function. Additional employment of a variable

factor input is desirable only when the marginal revenue productivity of that variable factor input is greater than or equal to the cost of employing it in an organization.

3. Production function also helps in making long run decisions. If returns to scale are increasing, it is wise to employ more factor units and increase production. If returns to scale are diminishing, it is unwise to employ more factor inputs & increase production. Managers will be indifferent whether to increase or decrease production, if production is subject to constant returns to scale.

Thus, production function helps both in the short run and long run decision-making process.

3. PRODUCTION FUNCTION WITH ONE VARIABLE INPUT

In this section, we will discuss the concept of production function with single variable input. This concept can be elaborated through the law of variable proportions.

The law of variable proportions

This law is one of the most fundamental laws of production. It gives us some key insights in determination of the ideal combination of factor inputs. All factor inputs are not available in plenty. Hence, in order to expand the output, scarce factors must be kept constant and variable factors are to be increased in greater quantities. Additional units of a variable factor on the fixed factors will certainly mean a variation in output. The law of variable proportions or the law of non-proportional output will explain how variation in one factor input leads to variations in output.

The law can be stated as follows: "As more and more units of a variable factor are applied to the given quantity of a fixed factor, total product may increase at an increasing rate initially, but eventually it will increase at a diminishing rate".

The law of variable proportions is an extension of the famous "law of diminishing returns" of classical economists. This law is stated by various economists. According to Prof. Benham, "As the proportion of one factor in a combination of factors is increased, after a point, first the marginal and then the average product of that factor will diminish".

The same idea has been expressed by Prof. Marshall in the following words: "An increase in the quantity of a variable factor added to fixed factors, at the end results in a less than proportionate increase in the amount of product, given technical conditions".

Assumptions of the law

Some assumptions of the law of variable proportions are as follows:

- Only one variable factor unit is to be varied while all other factors should be kept constant.
- All the units of a variable factor are homogeneous and are equally efficient.
- State of Technology is given and remains constant.
- The law will hold good only for a short period.

Let us see an illustration for better understanding.

Illustration

A hypothetical production schedule is worked out to explain the operation of the law. Table 5.1 shows the hypothetical production schedule.

Fixed factors = 1 Acre of land + Rs 500000 capital. Variable factor = labour.

Table 1: Hypothetical Production Schedule

Units of Variable inputs (Labour)	TP in units	AP in units	MP in units	
0	0	0	0	
1	10	10	10	I Stage
2	24	12	14	
3	39	13	15	
4	52	13	13	
5	60	12	8	
6	66	11	6	II Stage
7	70	10	4	
8	72	9	2	
9	72	8	0	III Stage
10	70	7	-2	

Total product or output (TP): It is the output derived from all units of factors, both fixed & variable, employed by the producer. In other words, it can also be define as the sum of marginal output.

Average product or output (AP): It can be obtained by dividing total output by the number of variable factors employed. In other words, average product is the ratio of total quantity produced and the amount of labor employed (variable factor) along with the given quantity of capital (fixed factor).

Marginal product or output (MP): It is the output derived from the employment of an additional unit of a variable factor.

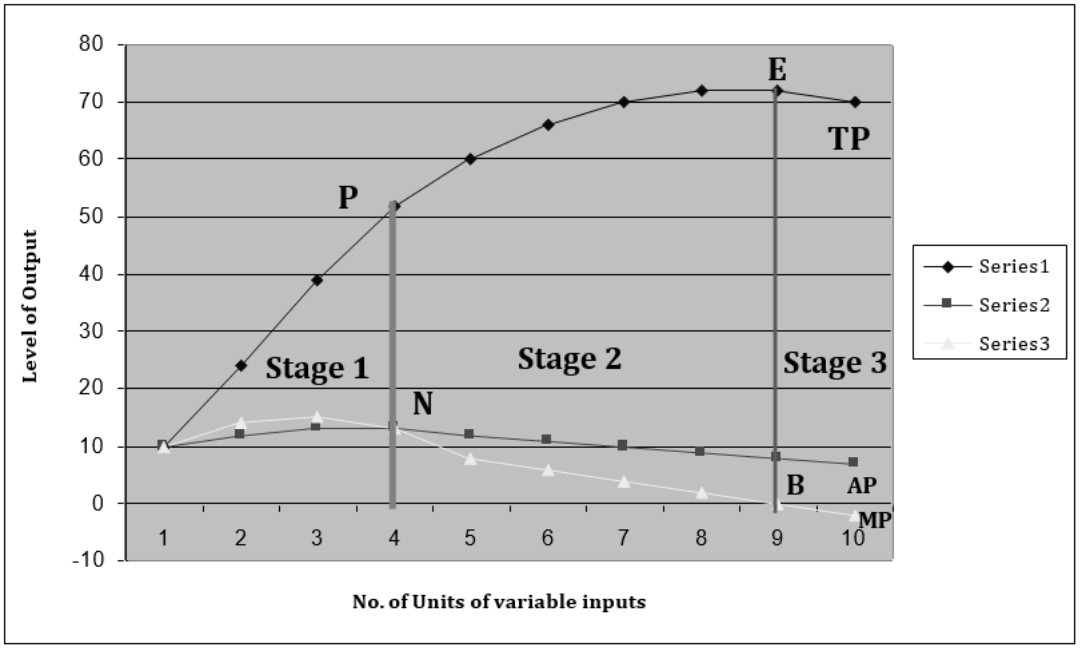
Trends in output

From table 5.1, one can observe the following tendencies in the TP, AP, & MP:

1. Total output goes on increasing as long as MP is positive. It is the highest when the MP is zero and the TP declines when the MP becomes negative.
2. MP increases in the beginning, reaches the highest point and diminishes at the end.

3. AP will also have the same tendencies as the MP. In the beginning MP will be higher than AP but at the end AP will be higher than MP.

Figure 2: depicts the diagrammatic representation of the output trends.



In figure 2, along the X axis, we measure the amount of variable factors employed and, along the Y axis; we measure TP, AP & MP. From the diagram it is clear that there are three stages.

Stage number I: Law of increasing returns

The total output increases at an increasing rate (More than proportionately) up to the point P because corresponding to this point P the MP is rising and reaches its highest point. After point P, MP decline and as such TP increases gradually. In figure 5.2, Point P does not correspond to maximum MP. The figure has to be redrawn.

The first stage comes to an end at the point where the MP curve cuts the AP curve when the AP is maximum at N.

The stage l is called the law of increasing returns on account of the following reasons:

1. The proportion of fixed factors is greater than the quantity of variable factors. When the producer increases the quantity of the variable factor, intensive and effective utilization of fixed factors becomes possible leading to higher output.
2. When the producer increases the quantity of the variable factor, output increases due to the complete utilization of the “indivisible factors”.

3. As more units of the variable factor are employed, the efficiency of variable factors will go up because it creates more opportunity for the introduction of division of labor and specialization thereby resulting in higher output.

Stage number II: Law of diminishing returns

In this case, as the quantity of variable inputs is increased to a given quantity of fixed factors, output increases less than proportionately. In this stage, the TP increases at a diminishing rate as both AP & MP are declining but, they are positive. The II stage comes to an end at the point where TP is the highest at point E and MP is zero at point B. It is known as the stage of "diminishing returns" because both the AP & MP of the variable factor continuously fall during this stage. It is only at this stage; the firm is maximizing its total output.

Diminishing returns arise due to the following reasons:

1. The proportion of variable factors is greater than the quantity of fixed factors. Hence, both AP & MP decline.
2. Total output diminishes because there is a limit to the full utilization of indivisible factors and introduction of specialization. Hence, output declines.
3. Diseconomies of scale will operate beyond the stage of optimum production.
4. Imperfect substitutability of factor inputs is another cause. Up to a certain point, substitution is beneficial. Once the optimum point is reached, the fixed factors cannot be compensated by the variable factor. Diminishing returns are bound to appear as long as one or more factors are fixed and cannot be substituted by the others.

Stage number III: Stage of negative returns

In this case, as the quantity of variable input is increased to a given quantity of fixed factors, output becomes negative. During this stage, TP starts diminishing, AP continues to diminish and MP becomes negative. The negative returns are the result of excessive quantity of variable factors to a constant quantity of fixed factors. Hence, output declines. Generally, the III stage is a theoretical possibility because no producer would like to reach this stage.

The producer, being rational, will not select either stage I (because there is an opportunity for him to increase output by employing more units of variable factor) or stage III (because the MP is negative). Stages I & III are described as a Non-Economic Region or Uneconomic Region. Hence, the producer will select the II stage (which is described as the most economic region) where he can maximize the

output. The II stage represents the range of rational production decisions.

It is clear that in the example, the most ideal or optimum combination of factor units is 1 Acre of land + Rs. 500000 capital and 9 laborers.

All the 3 stages together constitute the law of variable proportions. Because the second stage is the most important, in practice we normally refer to this law as the law of diminishing returns.

Practical application of the law

1. It helps a producer to work out the most ideal combination of factor inputs or the least cost combination of factor inputs.
2. It is useful to a businessman in the short run production planning at the micro-level.
3. The law gives guidance that by making continuous improvements in science and technology, the producer can postpone the occurrence of diminishing returns.

Thus, we have discussed the concept of production function with single variable input through the law of variable proportions.

4. PRODUCTION FUNCTION WITH TWO VARIABLE INPUTS

In this section we will discuss the concept of production function with two variable inputs.

Isoquants and Iso costs

The prime concern of a firm is to use the cheapest factor combinations to produce a given quantity of output. There are a large number of alternative combinations of factor inputs which can produce a given quantity of output for a given amount of investment. Hence, a producer has to select the most economical combination out of them. The Iso product curve is a technique developed in recent years to show the equilibrium of a producer with two variable factor inputs. It is a parallel concept to the indifference curve in the theory of consumption.

Meaning and definitions

The term "isoquant" has been derived from 'Iso' meaning equal and 'Quant' meaning quantity. Hence, isoquant is also called equal product curve or product indifference curve or constant product curve. An Iso product curve represents all the possible combinations of two factor inputs which are capable of producing the same level of output. It may be defined as – "a curve which shows the different combinations of the two inputs producing the same level of output".

Each isoquant curve represents only one particular level of output. If there are different isoquant curves, they represent different levels of output. Any point on an isoquant curve represents the same level of output. Because

Each point indicates an equal level of output, the producer becomes indifferent with respect to any one of the input combinations.

Equal product combination

Table 2: Schedule – 'Isoquant Schedule'

Combinations	Factor X(Labour)	Factor YCapital	Total Output inunits
A	12	1	100
B	8	2	100
C	5	3	100
D	3	4	100
E	2	5	100

In table 2, all the five factor combinations will produce an equal level of output, i.e.,100 units. Hence, the producer is indifferent with respect to any one of the factor combinations mentioned in the table.

Graphic representation

Figure 5.3 depicts the graphical representation of factor combination.

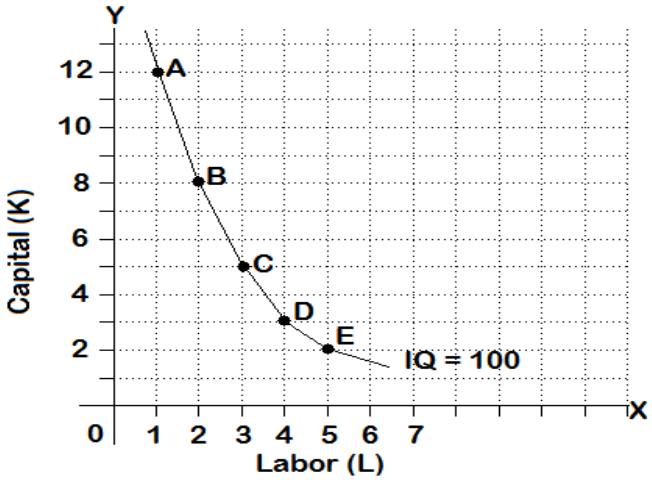


Figure 3: Graphical Representation of Factor Combination

In figure 5.3, if we join points ABCDE (which represents different combinations of factors x and y yielding the same level of output) we get an isoquant curve IQ. This curve represents 100 units of output that may be produced by employing any one of the combinations of two factor inputs as mentioned in the example. It is to be noted that an Iso product curve shows the exact physical units of output that can be produced by alternative combinations of two factor inputs. Hence, absolute measurement of output is possible.

Isoquant map

A catalogue of different combinations of inputs with different levels of output can be indicated in a graph which is called equal product map or isoquant map. In other words, a number of isoquants

representing different amounts of output are known as isoquant maps. Figure 5.4 depicts an isoquant map.

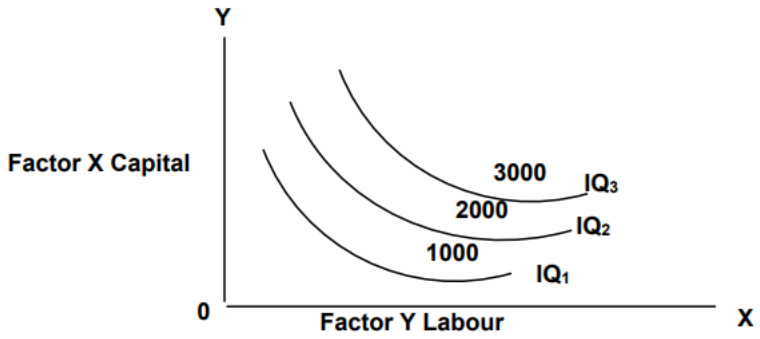


Figure 4: Isoquant Map

Marginal rate of technical substitution (MRTS)

It may be defined as the rate at which a factor of production can be substituted for another at the margin without affecting any change in the quantity of output. For example, MRTS of X for Y is the number of units of factor Y that can be replaced by one unit of factor X, quantity of output remaining the same.

Table 5.3 shows the MRTS of X for Y, for five different combinations.

Table 3: MRTS of X for Y

Combinations	Factor X	Factor Y	MRTS of X for Y
A	12	1	Nil
B	8	2	4:1
C	5	3	3:1
D	3	4	2:1
E	2	5	1:1

In this example, we can notice that in the second combination the producer is substituting 4 units of X for 1 unit of Y. Hence, in this case MRTS of X for Y is 4:1.

Generally speaking, the MRTS will be diminishing. In table 5.3, we can observe that as the quantity of factor Y is increased relative to the quantity of X, the number of units of X that will be required to be replaced by one unit of factor Y will diminish, quantity of output remaining the same. This is known as the law of diminishing marginal rate of technical substitution (DMRTS).

The properties of iso-quants are as follows:

1. An isoquant curve slope downwards from left to right.
2. Generally, an isoquant curve is convex to the origin.
3. No two Iso product curves intersect each other.
4. An Iso product curve lying to the right represents higher output and vice- versa.
5. Always, an isoquant curve need not be parallel to other.
6. Isoquant will not touch either X axis or Y axis.

Thus, we have learnt about MRTS, DMRTS and the isoquant.

Iso cost line or curve

It is a parallel concept to the budget or price line of the consumer. It indicates the different combinations of the two inputs which the firm can purchase at given prices with a given outlay. It shows two things: (a) prices of two inputs (b) total outlay of the firm. Each Iso cost line will show various combinations of two factors which can be purchased with a given amount of money at the given price of each input. We can draw the Iso cost line on the basis of an imaginary example.

Let us suppose that a producer wants to spend Rs. 3,000 to purchase factor X and Y. If the price of X per unit Rs. 100 he can purchase 30 units of X. Similarly, if the price of factor Y is Rs. 50 then he can purchase 60 units of Y. When 30 units of factor X are represented on OY – axis and 60 units of factor Y are represented on OX- axis, we get two points A & B. If we join these two points A and B, then we get the Iso cost line AB. This line represents the different combinations of factor X and Y that can be purchased with Rs. 3,000. Figure 5.5 depicts the Iso cost line.

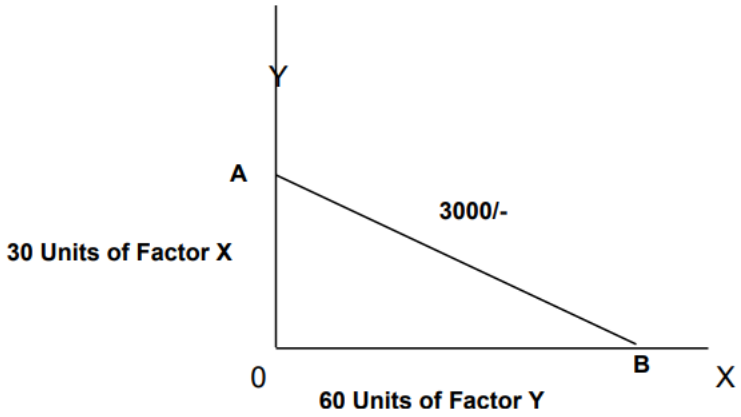


Figure 5: Isocost Line

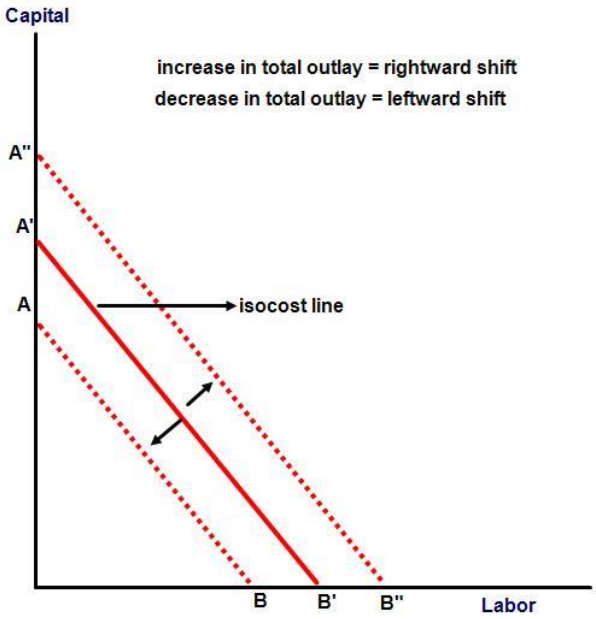


Figure 6: Effect of Changes in Factor on Iso cost Line

The Iso cost line will shift to the right if the producer increases the outlay from Rs. 3,000 to Rs. 4,000 that is from point AIBI to AII BII. On the contrary, if the outlay decreases to Rs. 2,000, there will be a backward shift in the position of Iso cost line that is from point AI BI to AB.

The slope of the Iso cost line represents the ratio of the price of a unit of factor X to the price of a unit of factor Y. In case, the price of any one of them changes, there would be a corresponding change in the slope and position of Iso cost line. Figure 5.6 depicts the change in the slope of the Iso cost line with changes in any of the factors.

Producer's equilibrium (Optimum factor combination or least cost combination)

The optimal combination of factor inputs may help in either minimizing cost for a given level of output or maximizing output with a given amount of investment expenditure (outlay). In order to explain producer's equilibrium, we have to integrate isoquant curve with the Iso cost line. The isoproduct curve represents different alternative possible combinations of two factor inputs with the help of which, a given level of output can be produced. On the other hand, Iso cost line shows the total outlay of the producer and the prices of factors of production.

The intention of the producer is to maximize the profits. Profits can be maximized when maximum output is produced with minimum production cost. Hence, the producer selects the least cost combination of the factor inputs. Maximum output with minimum cost is possible only when the

position of equilibrium is reached. The position of equilibrium is indicated at the point where the isoquant curve is tangent to the Iso cost line. Figure 5.7 explains how the producer reaches the position of equilibrium.

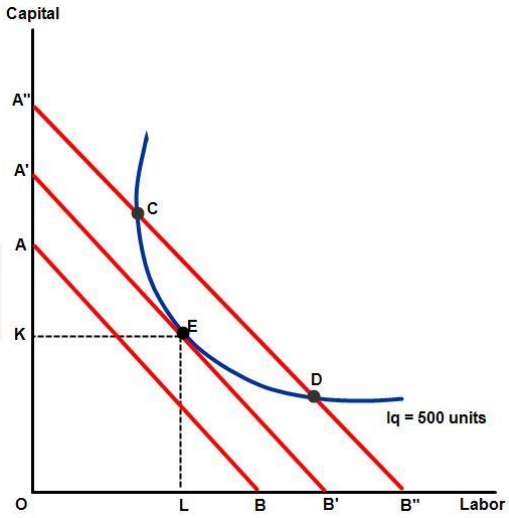


Figure 7: Producer's Equilibrium

It is quite clear from figure 5.7 that the producer will reach the position of equilibrium at the point E where the isoquant curve IQ (for output of 500 units) and Iso cost line A'B'are at a tangent to each other. With a given total outlay of Rs. 5,000 the producer will be producing the highest output,

i.e., 500 units by employing 25 units of factors X and 50 units of factor Y (assuming Rs. 2,500 each is spent on X and Y).

The price of one unit of factor X is Rs.100 and that of Y is Rs. 50; Rs.100 x 25 units of X = 2500 and Rs. 50 x 50 units of Y = 2500. The producer will not reach the position of equilibrium either at point C and D because they are on a higher Iso cost line A"B". Similarly, the producer cannot move to the left side of E, because they are on a lower Iso cost line and 500 units of output cannot be produced by any combination that lies to the left of E which is AB iso cost line.

Thus, the point at which the isoquant is tangent to the Iso cost line represents the minimum cost or optimum factor combination for producing a given level of output. At this point, MRTS between the two points is equal to the ratio of the prices of the inputs.

5. RETURNS TO SCALE

In this section, we will discuss the long run production function.

Laws of returns to scale

The concept of returns to scale is a long run phenomenon. In this case, we study the change in output when all factor inputs are changed or made available in the required quantity. An increase in scale means that all factor inputs are increased in the same proportion. In returns to scale, all the necessary factor inputs are increased or decreased to the same extent so that whatever the scale of production, the proportion among the factors remains the same.

Three phases of returns to scale

Generally speaking, we study the behavior pattern of output when all factor inputs are increased in the same proportion under returns to scale. Many economists have questioned the validity of returns to scale on the ground that all factor inputs cannot be increased in the same proportion and the proportion between the factor inputs cannot be kept uniform. But in some cases, it is possible that all factor inputs can be changed in the same proportion and the output is studied when the input is doubled or tripled or increased five-fold or ten-fold. An ordinary person may think that when the quantity of inputs is increased 10 times, output will also go up by 10 times. But it may or may not happen as expected.

It may be noted that when the quantity of inputs is increased in the same proportion, the scale of output or returns to scale may be either more than equal, equal or less than equal. Thus, when the scale of output is increased, we may get increasing returns, constant returns or diminishing returns.

When the quantity of all factor inputs is increased in a given proportion and output increases more than proportionately, then the returns to scale are said to be increasing; when the output increases in the same proportion, then the returns to scale are said to be constant; when the output increases less than proportionately, then the returns to scale are said to be diminishing. Table 5.4 shows an example to explain the law of returns to scale.

Table 4: Laws of Returns to Scale

Sl. No.	Scale	Total Productin Units	Marginal Productin units

1	1 Acre of land + 3 labor	5	5
2	2 Acre of land + 5 labor	12	7
3	3 Acre of land + 7 labor	21	9
4	4 Acre of land + 9 labor	32	11
5	5 Acre of land + 11 labor	43	11
6	6 Acre of land + 13 labor	54	11
7	7 Acre of land + 15 labor	63	9
8	8 Acre of land + 17 labor	70	7

It is clear from the table that the quantity of land and labor (scale) is increasing in the same proportion, i.e., by 1 acre of land and 3 units of labor throughout in our example. The output increases more than proportionately up to the point where the producer is employing 4 acres of land and 9 units of labor. Output increases in the same proportion when the quantity of land is 5 acres and 11 units of labor and 6 acres of land and 13 units of labor. In the later stages, when the producer employs 7 & 8 acres of land and 15 & 17 units of labor, output increases less than proportionately. Thus, one can clearly understand the operation of the three phases of the laws of returns to scale. First phase it increases, second phase output remains the same and then third phase the output gradually decreases.

Diagrammatic representation

Figure 5.7 depicts the marginal returns curve. In the figure, it is clear that the marginal returns curve slope upwards indicating increasing returns to scale. The curve is next almost parallel to X axis indicating constant but diminishing returns to scale, the curve slope downwards from left to right indicating the negative returns to scale.

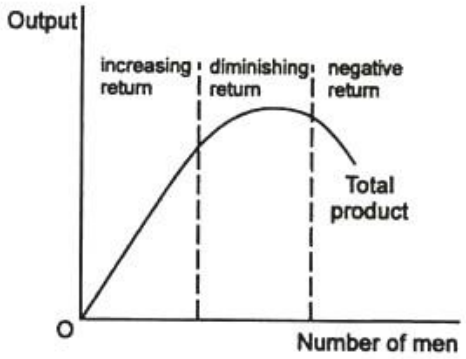


Figure 8: Marginal Returns Curve

Increasing returns to scale

Increasing returns to scale are said to operate when the producer is increasing the quantity of all factors [scale] in a given proportion leading to a more than proportionate increase in output. For example, when the quantity of all inputs is increased by 10%, and output increases by 15%, then we say that increasing returns to scale is operating. In order to explain the operation of this law, an equal product map has been drawn with the assumption that only two factors X and Y are required. Figure 5.9 depicts the operation of the law of increasing returns to scale. In the figure, Factor X is represented along OX axis and factor Y is represented along OY axis. The scale line OP is a straight line passing through the origin on the isoquant map indicating the increase in scale as we move upward. The scale line OP represents different quantities of inputs where the proportion between factor X and factor Y remains constant. When the scale is increased from A to B, the return increases the output from 100 units to 200 units. The scale line OP passing through origin is called the "expansion path". Any line passing through the origin will indicate the path of expansion or increase in scale with definite proportion between the two factors. It is very clear that the increase in the quantities of factor X and Y [scale] is small as we go up the scale and the output is larger. The distance between each isoquant curve is progressively diminishing. It implies that in order to get an increase in output by another 100 units, a producer is employing lesser quantities of inputs and his production cost is declining. Thus, the law of increasing returns to scale is operating.

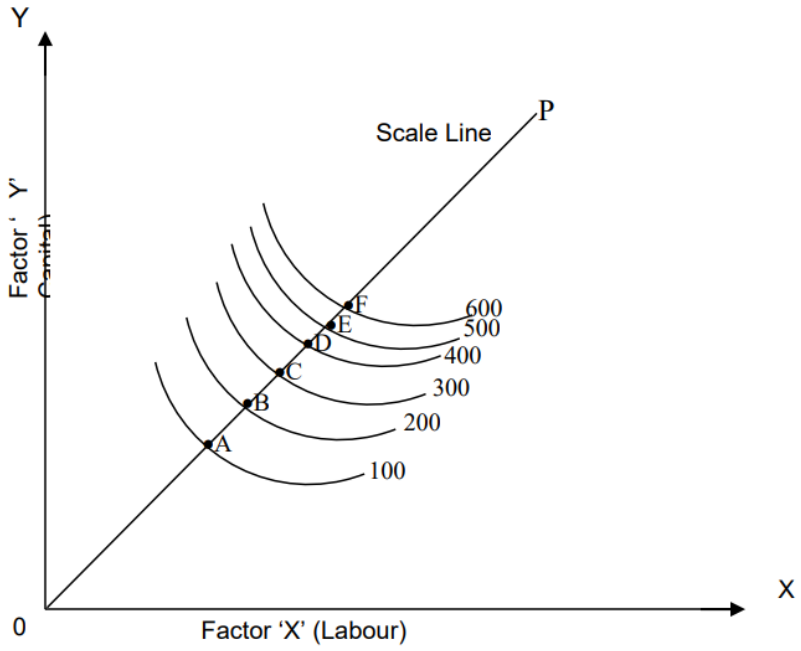


Figure 9: Operation of Increasing Returns of Scale

Causes for increasing returns to scale

Increasing returns to scale operate in a firm on account of several reasons. Some of the most important ones are as follows:

1. Wider scope for the use of the latest tools, equipment, machinery, techniques etc. to increase production and reduce cost per unit.
2. Large-scale production leads to full and complete utilization of indivisible factor inputs leading to further reduction in production cost.
3. As the size of the plant increases, more output can be obtained at a lower cost.
4. As output increases, it is possible to introduce the principle of division of labor and specialization, effective supervision and scientific management of the firm etc. leading to reduction in average cost of operations.
5. As output increases, it becomes possible to enjoy several other kinds of economies of scale like overhead, financial, marketing and risk-bearing economies, etc., which are responsible for cost reduction.

It is important to note that economies of scale outweigh diseconomies of scale in case of increasing returns to scale.

Constant returns to scale

Constant returns to scale are operating when all factor inputs [scale] are increased in a given proportion leading to an Equi-proportional increase in output. When the quantity of all inputs is increased by 10%, and output also increases exactly by 10%, then we say that constant returns to scale are operating. Figure 5.10 depicts a graph for constant returns to scale. In the figure, it is clear that the successive isoquant curves are equidistant from each other along the scale line OP. It indicates that as the producer increases the quantity of both factors X and Y in a given proportion, output also increases in the same proportion. Economists also describe constant returns to scale as the linear homogeneous production function. It shows that with constant returns to scale, there will be one input proportion which does not change, whatever the level of output.

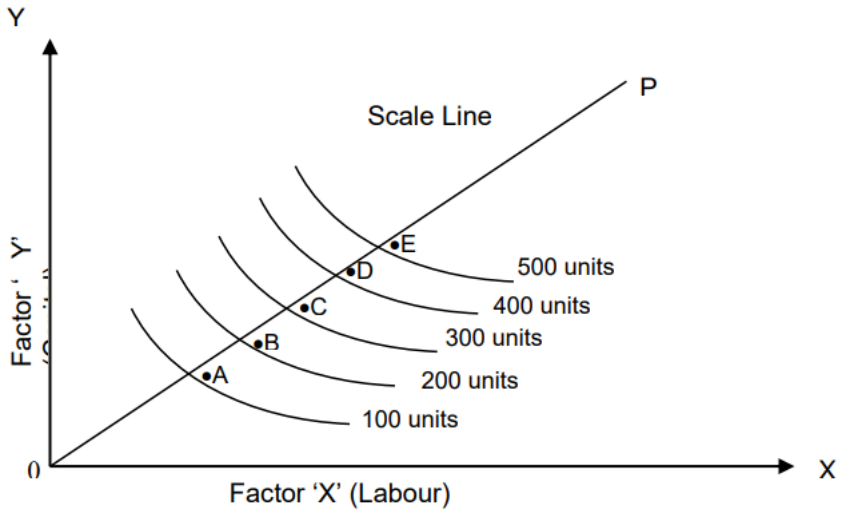


Figure 10: Constant Returns to Scale

Causes for constant returns to scale

In case of constant returns to scale, the various internal and external economies of scale are neutralized by internal and external diseconomies. Thus, when both internal and external economies and diseconomies are exactly balanced with each other, constant returns to scale will operate.

Diminishing returns to scale

Diminishing returns to scale is operating when output increases less than proportionately when compared to the quantity of inputs used in the production process. For example, when the quantity of all inputs is increased by 10%, and output increases by 5%, then we say that diminishing returns to scale is operating.

Figure 5.11 depicts the diminishing return to scales. In the figure, it is clear that the distance between each successive isoquant curve is progressively increasing along the scale line OP. It indicates that as the producer is increasing the quantity of both factors X and Y, in a given proportion, output increases less than proportionately. Thus, the law of diminishing returns to scale is operating.

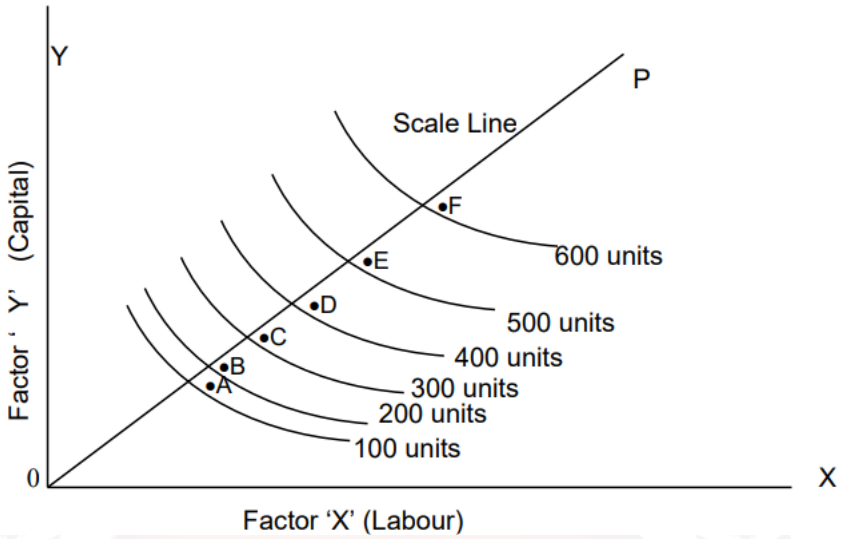


Figure 11: Diminishing Returns to Scale

Causes for diminishing returns to scale

Diminishing Returns to Scale operate due to the following reasons:

1. Emergence of difficulties in co-ordination and control.
2. Difficulty in effective and better supervision.
3. Delays in management decisions.
4. Inefficiency and mismanagement due to overgrowth and expansion of the firm.
5. Productivity and efficiency declining unavoidably after a point.

Thus, in this case, diseconomies outweigh economies of scale. The result is the operation of diminishing returns to scale.

The concept of returns to scale helps a producer to work out the most desirable combination of factor inputs so as to maximize output and minimize production cost. It also helps the producer to increase production, maintain the same level or decrease it depending on the demand for the product.

Cost Of Production

The cost of production refers to the total expense incurred by a firm in the process of producing goods or services. This concept encompasses various types of costs that can influence a firm's decisions in both the short and long run, impacting profitability and efficiency.

Types of Costs

1. Fixed Costs: These costs remain constant regardless of the level of production. Fixed costs include expenses like rent, salaries of permanent staff, and depreciation of machinery. They are incurred even when production is zero, making them significant for cost control in the short run.

2. Variable Costs: Variable costs fluctuate with the level of output. They include costs for raw materials, direct labour, and utilities such as electricity and water, which vary depending on the production volume. As production increases, variable costs also increase proportionately.

3. Total Cost: This is the sum of fixed and variable costs at any given production level. In equation form, it can be expressed as:

Total Cost (TC)=Fixed Cost (FC)+Variable Cost (VC)

4. Average Cost (AC): Average cost, or cost per unit, is obtained by dividing the total cost by the total quantity produced. It helps managers assess the efficiency of production. Average cost can further be divided into Average Fixed Cost (AFC) and Average Variable Cost (AVC), therefore, AC can also be calculated as the sum of AFC and AVC.

5. Marginal Cost: Marginal cost is the additional cost incurred in producing one more unit of output. This cost is crucial for decision-making, as it helps in determining the optimal level of production where profitability is maximized. The marginal cost curve typically declines initially, reaches a minimum, and then rises as production increases.

Importance of Cost Analysis

Cost analysis is a crucial tool for business decision-making, enabling firms to evaluate and manage the expenses associated with production. By examining both fixed and variable costs, cost analysis provides insights into how resources are allocated and used within an organisation, which directly affects profitability and operational efficiency.

Firstly, cost analysis helps firms identify areas where expenses can be optimised or reduced. By scrutinising each component of production costs, from raw materials to labour and overheads, managers can uncover inefficiencies and implement strategies to streamline operations, resulting in cost savings. For example, better inventory management can reduce storage costs, while negotiating supplier contracts may lower raw material expenses.

Secondly, cost analysis supports pricing decisions. By understanding the total cost of production, including both fixed and variable costs, firms can set prices that cover these costs while remaining competitive. Accurate pricing helps businesses avoid underpricing (which may lead to losses) and overpricing (which may drive customers away), thereby safeguarding profit margins.

In strategic planning, cost analysis also plays a key role in decisions about scaling production. In the long run, firms can analyse how costs vary with production levels, assessing whether economies of scale can be achieved by increasing output. This information is essential for capital investment decisions, such as expanding facilities or automating production processes, where understanding cost implications is fundamental to achieving a return on investment.

Moreover, cost analysis is integral to risk management. By forecasting costs under different scenarios, firms can prepare for market fluctuations, such as price changes in raw materials or shifts in demand, ensuring financial stability even in volatile conditions.

In sum, cost analysis enables businesses to make informed choices, improving cost-efficiency, profitability, and competitiveness in the marketplace.

Application in Production Function

The concept of cost of production is closely tied to the production function, which represents the relationship between input factors (such as labour, capital, and raw materials) and output levels. By analysing the cost of production, firms can determine the most efficient input combinations to produce a given output, thereby optimising both resource allocation and operational costs.

In a production function, firms seek to minimise costs while maximising output. This involves understanding the impact of different inputs on the overall production costs. For instance, in the short run, where at least one input is fixed, firms can vary their use of variable inputs (like labour and raw materials) to achieve the most cost-effective output level. This analysis enables firms to apply the "law of diminishing returns" effectively—understanding that, beyond a certain point, adding more of a variable input will yield smaller increases in output, thus increasing the marginal cost.

In the long run, where all inputs are variable, cost analysis within the production function aids in decisions regarding scaling operations. Managers can examine how changes in input levels, such as expanding production facilities or automating processes, influence costs and productivity. Through such analysis, firms can determine if they are experiencing increasing, constant, or diminishing returns to scale, which ultimately informs long-term strategic planning.

Moreover, understanding costs in relation to the production function helps in pricing decisions. If a firm knows the marginal cost of producing an additional unit, it can set prices that cover these costs and yield profit. Overall, cost analysis within the production function is vital for effective decision-making, enabling firms to strike a balance between cost control and output maximisation, enhancing competitive advantage in the market.

6. ECONOMIES OF SCALE

In this section, we will study the economies of scale. The study of economies of scale is associated with large-scale production. Today there is a general tendency to organise production on a large scale. Mass production of standardised goods has become the order of the day. Large-scale production is beneficial and economical in nature. The advantages or benefits that accrue to a firm as a result of an increase in its scale of production are called 'economies of scale'. They have a close relationship with the size of the firm. They influence the average cost over different ranges of output. They are beneficial to a firm. They help to reduce production costs and establish an optimum size for a firm. Thus, they help a lot and go a long way in the development and growth of a firm. According to Prof. Marshall, such economies are of two types, viz., internal economies and external economies. Now we shall study both of them in detail.

I. Internal economies or real economies

Internal economies are those economies which arise because of the actions of an individual firm to economize its cost. They arise due to increased division of labor or specialization and complete utilization of indivisible factor inputs. Prof. Cairncross points out that the internal economies are open to a single factory or a single firm, independent of the actions of other firms. Some of the important aspects of internal economies are as follows:

1. They arise "within" or "inside" a firm.
2. They arise due to improvements in internal factors.
3. They arise due to the specific efforts of one firm.
4. They are particular to a firm and enjoyed by only one firm.
5. They arise due to an increase in the scale of production.
6. They are dependent on the size of the firm.
7. They can be effectively controlled by the management of a firm.
8. They are called "business secrets" of a firm.

Internal economies arise on account of an increase in the scale of output of a firm and cannot be achieved unless output increases.

Kinds of internal economies

Now, let us discuss the kinds of internal economies.

Technical economies

These economies arise on account of technological improvements and their practical application in the field of business. Economies of techniques or technical economies are further subdivided into five heads as follows:

a) ***Economies of superior techniques*** **–** These economies are the result of the application of the most modern techniques of production. When the size of the firm grows, it becomes possible to employ bigger and better types of machinery. The latest and improved techniques give place for specialized production. It is bound to be cost-reducing in nature, for example, cultivating the land with modern tractors instead of using age-old wooden ploughs and bullock carts, using computers instead of human labour, etc.

b) ***Economies of increased dimension*** **–** It is found that a firm enjoys a reduction in cost when it increases its dimension. A large firm avoids wastage of time and economizes its expenditure. Thus, an increase in a firm's dimensions will reduce production costs, for example, the operation of a double-decker bus instead of two separate buses.

c) ***Economies of linked processes*** **–** It is quite possible that a firm may not have various processes of production within its own premises. Also, it is possible that different firms, through mutual agreement, may decide to work together and derive the benefits of linked processes, for example, in dairy farming, printing press, nursing homes, etc.

d) ***Economies arising out of research and by-products*** **–** A firm can invest adequate funds in research, and the benefits of research and its costs can be shared by that firm as well as all other firms in the industry. Similarly, a large firm can make use of its waste and by-products in the most economical manner by producing other products. For example, cane pulp, molasses, and bagasse of sugar factory can be used for the production of paper, varnish, etc.

e) ***Inventory economies*** **–** Inventory management is a part of better materials management. A big firm can save a lot of money by adopting the latest inventory management techniques. For example, Just-In-Time or zero level inventory techniques. The rationale of the Just-In-Time technique is that instead of having huge stocks worth of lakhs and crores of rupees, it can ask the seller of the inputs to supply them just before the commencement of work in the production department each day.

Managerial Economies

They arise because of better, efficient, and scientific management of a firm. Such economies arise in two different ways as follows:

a) ***Delegation of details*** **–** The general manager of a firm cannot look after the working of all processes of production. In order to keep an eye on each production process he has to delegate some of his powers or functions to trained or specialised personnel and thus relieve himself for co-ordination, planning and executing the plans. This will enable him to bring about improvements in the production process and in bringing down the cost of production.

b) ***Functional specialisation*** **–** It is possible to secure economies of large-scale production by dividing the work of management into several separate departments. Each department is run by an expert and the rest of the work is left in the hands of specialists. This will ensure better and more efficient productive management with scientific business administration. This would lead to higher efficiency and a reduction in the cost of production.

Thus, managerial economies are mainly governed by proper delegation of details and by functional specialisation.

Marketing or commercial economies

These economies arise on account of buying and selling goods on a large-scale basis at favourable terms. A large firm can buy raw materials and other inputs in bulk at concessional rates. As the bargaining capacity of a big firm is much greater than that of small firms, it can get quantity discounts and rebates. In this way, economies may be secured in the purchase of different inputs.

A firm can reduce its selling costs also. A large firm can have its sales agency and channel. The firm can have a separate sales organization, a marketing department manned by experts who are well-versed in the art of pushing the products onto the market. It can follow an aggressive sales promotion policy to influence the decisions of the consumers.

Financial economies

They arise from advantages secured by a firm in mobilising huge financial resources. A large firm on account of its reputation, name and fame can mobilise huge funds from money market, capital market, and other private financial institutions at concessional interest rates. It can borrow from banks at relatively cheaper rates. It is also possible to have large overdrafts from banks. A large firm can float debentures and issue shares and get subscriptions from the general public. Another advantage will

be that the raw material suppliers, machine suppliers etc., are willing to supply materials and components at comparatively low prices, because they are likely to get bulk orders. Thus, a big firm has an edge over small firms in securing sufficient funds more easily and cheaply.

Labour economies

These economies arise as a result of employing skilled, trained, qualified and highly experienced persons by offering higher wages and salaries. As a firm expands, it can employ a large number of highly talented people and get the benefits of specialisation and division of labour. It can also impart training to the existing labour force in order to raise the skills, efficiency and productivity of workers. New schemes may be chalked out to speed up the work, conserve scarce resources, economise on expenditure and save labor time. It can provide better working conditions, promotional opportunities, restrooms, sports rooms, etc., and create facilities like a subsidised canteen, crèches for infants, and recreation. All these measures will definitely raise the average productivity of a worker and reduce the cost per unit of output.

Transport and storage economies

They arise on account of the provision of better, highly organised and cheap transport and storage facilities and their complete utilisation. A large company can have its own fleet of vehicles or means of transport which are more economical than hired ones. Similarly, a firm can also have its own storage facilities which reduce the cost of operations.

Overhead economies

These economies arise on account of large-scale operations. The expenses of establishment, administration, bookkeeping, etc., are more or less the same whether production is carried out on a small or large scale. Hence, the cost per unit will be low if production is organised on a large scale.

Economies of vertical integration

A firm can also reap this benefit when it succeeds in integrating a number of stages of production. It secures the advantage that the flow of goods through various stages in production processes is more readily controlled. Because of vertical integration, most of the costs become controllable costs which help an enterprise to reduce the cost of production.

Risk-bearing or survival economies

These economies arise as a result of avoiding or minimising several kinds of risks and uncertainties in a business. A manufacturing unit has to face a number of risks in the business. Unless these risks

are effectively tackled, the firm's survival may become difficult. Hence, many steps are taken by a firm to eliminate or to avoid or to minimise various kinds of risks. A large firm secures risk-spreading advantages in either of the following four ways or through all of them:

- ***Diversification of output*** **–** Instead of producing only one particular product, a firm has to produce multiple products. If there is a loss in one product, it can be made good in other products.
- ***Diversification of market*** **–** Instead of selling the goods in only one market, a firm has to sell its products in different markets. If consumers in one market desert a product, it can cover the losses in other markets.
- ***Diversification of source of supply*** **–** Instead of buying raw materials and other inputs from only one source, it is better to purchase them from different sources. If one person fails to supply, a firm can buy from several sources.
- ***Diversification of the process of manufacture*** **–** Instead of adopting only one process of production to manufacture a commodity, it is better to use different processes or methods to produce the same commodity so as to avoid the loss arising out of the failure of any one process.

Generally speaking, the risk-bearing capacity of a big firm will be much greater than that of a small firm. Risk is avoided when a few firms amalgamate or join together or when competition between different firms is either eliminated or reduced to a minimum by expanding the size of the firm.

II. External economies or pecuniary economies

External economies are those economies which accrue to the firms as a result of the expansion in the output of the whole industry and they are not dependent on the output level of individual firms. These economies or gains will arise on account of the overall growth of an industry or a region or a particular area. They arise due to the benefit of localisation and specialised progress in the industry or region. Some of the important aspects of external economies are as follows:

1. They occur 'outside' the firm.
2. They arise due to improvement in external factors.
3. They result out of collective efforts of an industry.
4. By nature, they are general, common and enjoyed by all firms.
5. They are caused due to overall development, expansion and growth of an industry or a region.
6. They are dependent on the size of industry.
7. They are beyond the control of management of a firm.

8. They are called “open secrets” of a firm.

Stonier & Hague point out that external economies are those economies in production which depend on an increase in the output of the whole industry rather than an increase in the output of the individual firm.

Kinds of external economies

Now, let us discuss the kinds of external economies.

Economies of concentration or agglomeration

They arise due to a very large number of firms which produce the same commodity being established in a particular area. In other words, this is an advantage which arises from what is called ‘Localization of Industry’. The following benefits of localisation of industry are enjoyed by all the firms - provision of better and cheap labour at low or reasonable rates, trained, educated and skilled labour, transport and communication, water, power, raw materials, financial assistance through private and public institutions at low interest rates, marketing facilities, benefits of common repairs, maintenance and service shops, services of specialists or outside experts, better use of by-products and other such benefits. Thus, it helps in reducing the cost of operation of a firm.

Economies of information

These economies arise as a result of getting quick, latest and up-to-date information from various sources. Another form of benefit that arises due to the localization of industry is economies of information. As a large number of firms are located in a region, it becomes possible for them to exchange their views frequently, to have discussions with others, to organize lectures, symposiums, seminars, workshops, training camps, demonstrations on topics of mutual interest, etc. Revolutions in the field of information technology, expansion in internet facilities, mobile phones, e-mails, video conferences, etc., have helped in the free flow of the latest information from all parts of the globe in a very short span of time. Similarly, publication of journals, magazines, information papers, etc. have helped in the dissemination of quick information. Statistical, technical and other market information becomes more readily available to all firms. This will help in developing contacts between different firms. When an inter-firm relationship strengthens, it helps in economizing the expenditure of a single firm.

Economies of disintegration

These economies arise as a result of dividing one big unit into different small units for the sake of convenience of management and administration. When an industry grows beyond a limit, in that case, it becomes necessary to split it into small units. New subsidiary units may grow to serve the needs of the main industry. For example, in the cotton textiles industry, some firms may specialize in manufacturing threads, a few others in printing, and some others in dyeing and coloring, etc. This will certainly enhance the efficiency in the working of a firm and cut down unit costs considerably.

Economies of government action

These economies arise as a result of the government's active support and assistance in stimulating production in the private sector. In recent years, the government has devised several kinds of assistance to encourage the development of private industries. It grants tax concessions, tax holidays, tax exemptions, subsidies, development rebates, financial assistance at low interest rates, etc.

This detailed description makes it clear that both internal and external economies arise due to large-scale production and benefit firms by reducing costs.

Economies of physical factors

These economies arise due to the availability of favourable physical factors and the environment. As the size of an industry expands, a positive physical environment may help to reduce the costs of all firms working in the industry. For example, climate, weather conditions, fertility of the soil, physical environment in a particular place may help all firms to enjoy certain physical benefits.

Economies of welfare

These economies arise on account of various welfare programs undertaken by an industry to help its own staff. A big industry is in a better position to provide welfare facilities to the workers. It may get land at concessional rates and procure special facilities from the local governments for setting up housing colonies for the workers. It may also establish health care units, training centers, computer centers and educational institutions of all types. It may grant concessions to its workers. All these measures would help in raising the overall efficiency and productivity of workers.

Diseconomies of scale

When a firm expands beyond the optimum limit, economies of scale will be converted into diseconomies of scale. Some of the main diseconomies of scale are as follows:

1. ***Financial diseconomies*** **–** As there is overgrowth, the required amount of finance may not be available to a firm. Consequently, higher interest rates are to be paid for additional funds.
2. ***Managerial diseconomies*** **–** Excess growth leads to a loss of effective supervision, control, management, and coordination of factors of production leading to all kinds of wastages, indiscipline and a rise in production and operating costs.
3. ***Marketing diseconomies*** **–** Unplanned excess production may lead to a mismatch between demand and supply of goods leading to a fall in prices. Stocks may pile up; sales may decline leading to a fall in revenue and profits.
4. ***Technical diseconomies*** **–** When output is carried beyond the plant capacity, per unit cost will certainly go up. There is a limit on the division of labour and specialization. Beyond a point, they become negative. Hence, operation costs would go up.
5. ***Diseconomies of risk and uncertainty bearing*** **–** If output expands beyond a limit, investment increases. The level of inventory goes up. Sales do not go up correspondingly. Business risks appear in all fields of activities. Supply of factor inputs becomes inelastic leading to high prices.
6. ***Labour diseconomies*** **–** An unwieldy firm may become impersonal. Contact between labor and management may disappear. Workers may demand higher wages and salaries, bonuses and other such benefits, etc. Industrial disputes may arise. Labour unions may not cooperate with the management. All of them may contribute to higher operation costs.

On account of diseconomies of scale, more output is obtained at higher cost of production.

III. External diseconomies

When several business units are concentrated in one place or locality, it may lead to congestion, environmental pollution, scarcity of factor inputs like raw materials, water, power, fuel, transport and communications etc. leading to higher production and operational costs.

Thus, it is very clear that a firm can enjoy the benefits of large-scale production only up to a limit. Beyond the optimum limit, it is bound to experience diseconomies of scale. Hence, there should be proper checks on the growth and expansion of a firm.

Internalization of external economies

It implies that a firm will convert certain external benefits created by the government or the entire society to its own favour without making any additional investments. A firm may start a new unit between two big railway stations or near the airport or near the national highways or a port so that it can enjoy all the infrastructure benefits. Similarly, a new computer firm can commence its

operations where there is 24 24-hour supply of electricity. Hence, they are also called privatization of public benefits. Such types of efforts are to be encouraged by the government.

Externalization of internal diseconomies

In this case, a particular firm, on account of its regular operations, will pass on certain costs to the entire society. A firm, instead of taking certain precautionary measures by spending some amount of money, will escape and pass on this burden to the government or society. For example, a firm may throw chemical or industrial waste, dirt and filth either into the open air or rivers leading to environmental pollution. In that case, the government is forced to spend more money to clean river water or prevent environmental pollution. This is a clear case of externalized internal diseconomies. It is to be avoided at all costs.

7. ECONOMIES OF SCOPE

In this section, we will discuss the economies of scope. It is a common fact to observe that when a single-product firm expands its volume of output, it would enjoy certain economies of scale. As a result, production cost per unit declines and more output is obtained at a lower cost of production. Sometimes, they would enjoy certain other external benefits due to the overall improvements in the entire area or city in which they operate. Apart from these two types of benefits, we also come across another type of benefit in recent years. They are popularly known as economies of scope.

Economies of scope may be defined as those benefits which arise to a firm when it produces more than one product jointly rather than producing two items separately by two different business units. In this case, the benefits of the joint output of a single firm are greater than the benefits if two products are produced separately by two different firms. Such benefits may arise on account of joint use of production facilities, joint marketing efforts, or use of the same administrative office and staff in an organization. Sometimes, production of one product automatically results in the production of another by-product leading to a reduction in the average cost of production.

Economies-of-scope results in saving production costs. It can be measured with the help of the following equation.

$$SC = \frac{C\,[Q1] + C\,[Q2] - C\,[Q1 + Q2]}{C\,[Q1 + Q2]}$$

Where SC = Saving Cost, C [Q1] = cost of producing output Q1, C [Q2] = cost of producing output Q2 and C [Q1+Q2] = joint cost of producing both outputs.

Illustration

A firm produces products A & B separately. The cost of producing 100 units of A is Rs. 8000 and the cost of producing 100 units of B is Rs. 5,000. If the firm produces both products A & B jointly, in that case, its total cost would be Rs. 10,000.

Now, one can find out saving cost by substituting the values to the above-mentioned formula.

$$SC = \frac{8{,}000 + 5{,}000 - 10{,}000}{10{,}000} = \frac{3{,}000}{10{,}000} = 0.3$$

In this case, the joint cost [10,000] is less than the sum of individual costs [13,000]. Thus, a firm can save 30% cost if it produces both products A & B jointly. Hence, the SC is more than zero.

Diseconomies of scope

Diseconomies of scope may be defined as those disadvantages which occur when the cost of producing two products jointly is costlier than producing them individually. In this case, it would be profitable to produce two goods separately than jointly. For example, with the help of the same machinery, it is not possible to produce two goods together. It involves buying two different machines. Hence, production costs would certainly go up in this case.

Difference between economies of scale and economies of scope

Table 5.5 shows the difference or the comparison between economies of scale and economies of scope.

Table 5: Comparison of Two Economies

Economies of scale	Economies of scope
1. It is connected with an increase ordecrease in the scale of production	1. It is connected with an increase or decrease in distribution & marketing.
2. It shows the change in output of asingle product	2. It shows a change in output of more than one product.
3. It is associated with SupplySidechanges in output.	3. It is associated with demand side changes in output
4. It indicates savings in cost owing to an increase in volume ofoutput	4. It indicates savings in cost due to production of more than one product.

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks:	
1	Production creates_____ or_____ of value.
2	Production function explains_____ or _____ relationship between inputs and outputs.
3	In the short period only_____ factor inputs are changed.
4	When marginal product is zero total product will be_____ .
5	An iso-quant curve shows different alternative combinations of inputs which helps to produce the same level of output whereas an iso-cost curve show _____ combination of two inputs that can be purchased with a given amount of investment expenditure while prices of two factor inputs remain constant.
6	When all inputs are increased by 8% and output increases by 13% then it is a case of law of _____.
7	Internal economies depend on the growth of a_____ and external economies depend on the growth of the_____
8	Economies of scope refer to the benefits which arise to a firm when it produces more than _____ separately by two firms _____ rather than producing.
True or False Questions:	
9	A short-run production function assumes that the level of output is fixed. (True/False)
10	When the average product is increasing, then the marginal product is greater than the average product. (True/False)
11	All inputs are fixed in the short run. (True/False)
12	If a firm is producing a given level of output in a technically efficient manner, that level of output is the most that can be produced with the given levels of inputs. (True/False)
13	Diminishing returns refer to the decrease in marginal product, which results from increases in the variable input. (True/False)

8. TYPES OF COSTS

In this section we will discuss the types of costs. The types of costs are as follows:

1. Money cost and real cost – When cost is expressed in terms of money, it is called money cost. It relates to money outlays by a firm on various factor inputs to produce a commodity. In a monetary economy, all kinds of cost estimations and calculations are made in terms of money only. Hence, the knowledge of money cost is of great importance in economics. Exact measurement of money cost is possible.

When cost is expressed in terms of physical or mental efforts put in by a person in the making of a product, it is called real cost. It refers to the physical, mental or psychological efforts, the exertions, sacrifices, the pains, the discomforts, displeasures and inconveniences which various members of society have to undergo to produce a commodity. It is a subjective and relative concept and hence exact measurement is not possible.

2. Implicit or imputed costs and explicit costs – Explicit costs are those costs which are in the nature of contractual payments and are paid by an entrepreneur to the factors of production [excluding himself] in the form of rent, wages, interest and profits, utility expenses, and payments for raw materials, etc. They can be estimated and calculated exactly and recorded in the books of accounts.

Implicit or imputed costs are implied costs. They do not take the form of cash outlays and as such do not appear in the books of accounts. They are the earnings of owner-employed resources. For example, the factor inputs owned by the entrepreneur himself like capital that can be utilized by him or can be supplied to others for a contractual sum. It is to be remembered that the total cost is a sum of both implicit and explicit costs.

3. Actual costs and opportunity costs – Actual costs are also called as outlay costs, absolute costs and acquisition costs. They are those costs that involve financial expenditures at some time and hence, are recorded in the books of accounts. They are the actual expenses incurred for producing or acquiring a commodity or service by a firm, for example, wages paid to workers, expenses on raw materials, power, fuel and other types of inputs. They can be exactly calculated and accounted without any difficulty.

Opportunity cost of a good or service is measured in terms of revenue which could have been earned by employing that good or service in some other alternative uses. In other words, the opportunity

cost of anything is the cost of displaced alternatives or costs of sacrificed alternatives. It implies that opportunity cost of anything is the alternative that has been foregone. Hence, they are also called alternative costs. Opportunity cost represents only sacrificed alternatives. Hence, they can never be exactly measured and recorded in the books of accounts.

The knowledge of opportunity cost is of great importance to managerial decision-making. The concept of opportunity cost helps in taking decisions to select the best alternative. While taking a decision among several alternatives, a manager selects the best one which is more profitable or beneficial by sacrificing other alternatives. For example, a firm may decide to buy a computer which can do the work of 10 laborers. If the cost of buying a computer is much lower than that of the total wages to be paid to the workers over a period of time, it will be a wise decision. On the other hand, if the total wage bill is much lower than that of the cost of the computer, it is better to employ workers instead of buying a computer. Thus, a firm has to take a number of decisions almost daily.

4. Direct costs and indirect costs – Direct costs are those costs which can be specifically attributed to a particular product, a department, or a process of production. For example, expenses on raw materials, fuel, wages to workers, salary to a divisional manager, etc. are direct costs. On the other hand, indirect costs are those costs which are not traceable to any one unit of operation. They cannot be attributed to a product, a department or a process. For example, expenses incurred on electricity bills, water bills, telephone bills, administrative expenses, etc.

5. Past and future costs – Past costs are those costs which are spent in the previous periods. On the other hand, future costs are those which are to be spent in the future.

6. Fixed costs and variable costs – Fixed costs are those costs which do not vary with either expansion or contraction in output. They remain constant irrespective of the level of output. They are positive even if there is no production. They are also called supplementary or overhead costs.

On the other hand, variable costs are those costs which directly and proportionately increase or decrease with the level of output produced; they are also called prime costs or direct costs.

7. Marginal and incremental costs – Marginal cost refers to the cost incurred on the production of another or one more unit. It implies the additional cost incurred to produce an additional unit of output.

Incremental cost on the other hand refers to the costs involved in the production of a batch or group of output. They are the added costs due to a change in the level or nature of business activity. For

example, the cost involved in the setting up of a new sales depot in another city or, the cost involved in the production of 100 extra units.

8. Accounting costs and economic costs – Accounting Costs are the actual, explicit costs incurred in the production of a commodity, which include acquisition costs (e.g., materials, labor, and other direct costs). These costs are recorded in financial statements and are usually associated with the production process. On the other hand, **Economic Costs**, on the other hand, are broader than accounting costs. They include both **explicit costs** (the same as accounting costs) and **implicit costs** (also known as opportunity costs). Implicit costs represent the value of resources in their next best alternative use. Economic costs are relevant for decision-making because they consider the trade-offs an entrepreneur faces when choosing between different options or courses of action. So, economic costs include the **total opportunity cost** of using resources in a particular way, whereas accounting costs focus only on the actual expenses incurred.

Determinants of costs

Cost behavior is the result of many factors and forces. But it is very difficult to determine in general the factors influencing costs, as they widely differ from firm to firm and even industry to industry. However, economists have given some factors considering them as general determinants of costs. They have enough importance in modern businesses and decision-making processes. The following factors deserve our attention in this connection:

1. **Technology –** Modern technology leads to optimum utilization of resources, avoidance of all kinds of wastages, saving of time, reduction in production costs and results in higher output. On the other hand, primitive technology would lead to higher production costs.
2. **Rate of output (degree of utilization of the plant and machinery) –** Complete and effective utilization of all kinds of plants and equipment would reduce production costs while, underutilization of existing plants and equipment would lead to higher production costs.
3. **Size of Plant and scale of production –** Big companies with huge plants and machinery organize production on large scale basis and enjoy economies of scale which reduce the cost per unit.
4. **Prices of factor inputs –** Higher market prices of various factor inputs result in higher cost of production and vice versa.
5. **Efficiency of factors of production and the management –** Higher productivity and efficiency of factors of production would lead to lower production costs and vice versa.

6. **Stability of output –** Stability in production would lead to optimum utilization of the existing capacity of plants and equipment. It also brings savings of various kinds of hidden costs of interruption and learning leading to higher output and reduction in production costs.
7. **Law of returns –** Increasing returns would reduce the cost of production and diminishing returns would increase costs.
8. **Time period –** In the short run, the cost will be relatively high and in the long run, it will be low as it is possible to undertake adjustments and readjustments in the production process.

Thus, many factors influence the cost of production of a firm.

9. COST-OUTPUT RELATIONSHIP: COST FUNCTION

In this section, we will discuss the cost-output relationship, i.e., the cost function. Cost and output are correlated. Cost-output relations play an important role in almost all business decisions. It throws light on cost minimization or profit maximization and optimization of output. The relation between the cost and output is technically described as the "cost function". The significance of cost-output relationship is so great that in economic analysis, the cost function usually refers to the relationship between cost and rate of output alone and we assume that all other independent variables are kept constant. Mathematically speaking TC = f (Q) where TC = Total cost and Q stands for output produced.

However, the cost function depends on three important variables. These variables are as follows:

1. **Production function –** If a firm is able to produce higher output with a small quantity of inputs, in that case, the cost function becomes cheaper and vice versa.
2. **The market prices of inputs –** If the market prices of different factor inputs are high in that case, cost function becomes higher and vice versa.
3. **Period of time –** Cost function becomes cheaper in the long run and it would be relatively costlier in the short run.

These variables govern the behavior of the cost function.

Types of cost functions

Generally speaking, there are two types of cost functions, namely, short run cost function and long run cost function.

10. COST-OUTPUT RELATIONSHIPS IN THE SHORT RUN

In this section, we will discuss the cost-output relationship in the short run. It is interesting to note that the relationship between the cost and output is different at two different periods of time i.e., short-run and long run. Generally speaking, the cost of production will be relatively higher in the short- run when compared to the long run. This is because a producer

will get enough time to make all kinds of adjustments in the productive process in the long run than in the short run. When cost and output relationship is represented with the help of diagrams, we get short run and long run cost curves of the firm. Now we shall make a detailed study of cost output relations both in the short run as well as in the long run.

Meaning of short run

The short run is a period of time in which only the variable factors can be varied while fixed factors like plant and machinery, etc. remain constant. Hence, the plant capacity is fixed in the short run. The total number of firms in an industry will remain the same. Time is insufficient either for the entry of new firms or exit of the old firms. If a firm wants to produce greater quantities of output, it can do so only by employing more units of variable factors or by having additional shifts, or by having overtime work for the existing labor force or by intensive utilization of existing stock of capital assets, etc. Hence, short run is defined as a period where adjustments to changed conditions are only partial.

The short run cost function relates to the short run production function. It implies two sets of input components: (a) fixed inputs and (b) variable inputs. Fixed inputs are unalterable. They remain unchanged over a period of time. On the other hand, variable factors are changed to vary the output in the short run. Thus, in the short period, some inputs are fixed in amount and a firm can expand or contract its output only by changing the amounts of other variable inputs. The cost-output relationship in the short run refers to a particular set of conditions where the scale of operations is limited by the fixed plant and equipment. Hence, the costs of the firm in the short run are divided into fixed costs and variable costs. We shall study these two concepts of costs in some detail.

Fixed costs

These costs are incurred on fixed factors like land, buildings, equipment, plants, superior type of labor, top management, etc. Fixed costs in the short run remain constant because the firm does not change the size of plant and the amount of fixed factors employed. Fixed costs do not vary with either

expansion or contraction in output. These costs are to be incurred by a firm even if output is zero. Even if the firm closes down its operations for some time temporarily in the short run, but remains in business, these costs have to be borne by it. Hence, these costs are independent of output and are referred to as unavoidable contractual costs.

Prof. Marshall called fixed costs supplementary costs. They include items such as contractual rent payment, interest on capital borrowed, insurance premiums, depreciation and maintenance allowances, administrative expenses like manager's salary or salary of the permanent staff, property and business taxes, license fees, etc. They are called as overhead costs because these costs are to be incurred whether there is production or not. These costs are to be distributed on each unit of output produced by a firm. Hence, they are called indirect costs.

Variable costs

The costs corresponding to variable factors are discussed as variable costs. These costs are incurred on raw materials, ordinary labor, transport, power, fuel, water, etc., which directly vary in the short run. Variable costs directly and proportionately increase or decrease with the level of output. If a firm shuts down for some time in the short run, then it will not use the variable factors of production and therefore will not incur any variable costs. Variable costs are incurred only when some amount of output is produced. Total variable costs increase with the increase in the level of production and vice versa. Prof. Marshall called variable costs, prime costs or direct costs because the volume of output produced by a firm depends directly upon them.

It is clear from the above description that production costs consist of both fixed as well as variable costs. The difference between the two is meaningful and relevant only in the short run. In the long run, all costs become variable because all factors of production become adjustable and variable in the long run.

However, the distinction between fixed and variable costs is very significant in the short run because it influences the average cost behavior of the firm. In the short run, even if a firm wants to close down its operations but wants to remain in business, it will have to incur fixed costs but it must cover at least its variable costs.

Cost-output relationship and nature and behavior of cost curves in the short run

In order to study the relationship between the level of output and corresponding cost of production, we have to prepare the cost schedule of the firm. A cost schedule is a statement of variations in costs

resulting from variations in the levels of output. It shows the response of costs to changes in output. Table 1 represents a hypothetical cost schedule of a firm.

Table 6: Hypothetical Cost Schedule

Output in Units	TFC in Rs.	TVC in Rs.	TC in Rs.	AFC in Rs.	AVC in Rs.	AC In Rs.	MC in Rs.
0	360	–	360	–	–	–	–
1	360	180	540	360	180	540	180
2	360	240	600	180	120	300	60
3	360	270	630	120	90	210	30
4	360	315	675	90	78.75	168.75	45
5	360	420	780	72	84	156	105
6	360	630	990	60	105	165	210

On the basis of the cost schedule, we can analyze the relationship between changes in the level of output and costs of production. If we represent the relationship between the two in a graphical manner, we get different types of cost curves in the short run.

In the short run, we will study the following kinds of cost concepts and cost curves.

Total fixed cost (TFC)

TFC refers to total money expenses incurred on fixed inputs like plant, machinery, tools and equipment in the short run. The total fixed cost corresponds to the fixed inputs in the short run production function. TFC remains the same at all levels of output in the short run. It is the same even when output is nil. It indicates that whatever the quantity of output may be, whether 1 to 6 units, TFC remains constant. Figure 1 depicts the total fixed cost curve. The TFC curve is horizontal and parallel to OX-axis, showing that it is constant regardless of output per unit of time. TFC starts from a point on Y-axis indicating that the total fixed cost will be incurred even if the output is zero. In our example, Rs. 360 is the TFC. It is obtained by summing up the product or quantities of the fixed factors multiplied by their respective unit price.

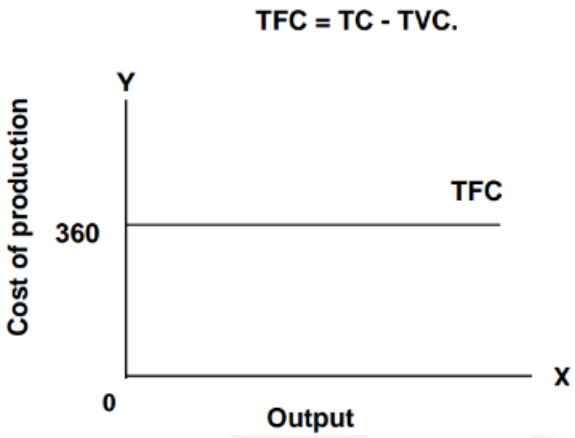


Figure 12: Total Fixed Cost Curve

Total Variable Cost (TVC)

TVC refers to total money expenses incurred on the variable factor inputs like raw materials, power, fuel, water, transport and communication, etc., in the short run. Total variable cost corresponds to variable inputs in the short run production function. It is obtained by summing up the quantities of variable inputs multiplied by their prices. The formula to calculate TVC is as follows. TVC = TC - TFC. TVC = f (Q), i.e., TVC is an increasing function of output. In other words, TVC varies with output. It is nil if there is no production. Thus, it is a direct cost of output. Figure 2 depicts the total variable cost curve. TVC rises sharply in the beginning, gradually in the middle and sharply at the end in accordance with the law of variable proportions. The law of variable proportions explains that in the initial stages, to obtain a given quantity of output, the relative change (variation) in variable factors that are needed are in lower proportion, but after a point when the diminishing returns operate, variable factors are to be employed in a larger proportion to increase the same level of output.

TVC curve slope upwards from left to right. TVC curve rises as output is expanded. When output is zero, TVC also will be zero. Hence, the TVC curve starts from the origin.

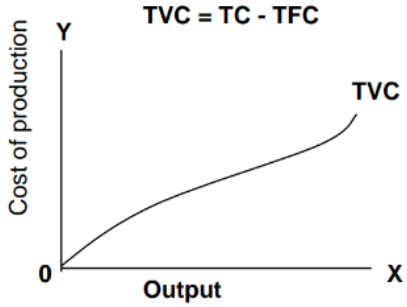


Figure 13: Total Variable Cost Curve

Total cost (TC)

Total cost refers to the aggregate expenditure incurred by a firm to produce a given quantity of output. The total cost is measured in relation to the production function by multiplying the factor quantities with their prices. TC = f (Q) which means that TC varies with output. Theoretically speaking, TC includes all kinds of money costs, both explicit and implicit costs. Normal profit is included in the total cost as it is an implicit cost. It includes fixed as well as variable costs. Hence, TC = TFC +TVC. Figure 3 depicts the total cost curve.

TC varies in the same proportion as TVC. In other words, a variation in TC is the result of variation in TVC since TFC is always constant in the short run.

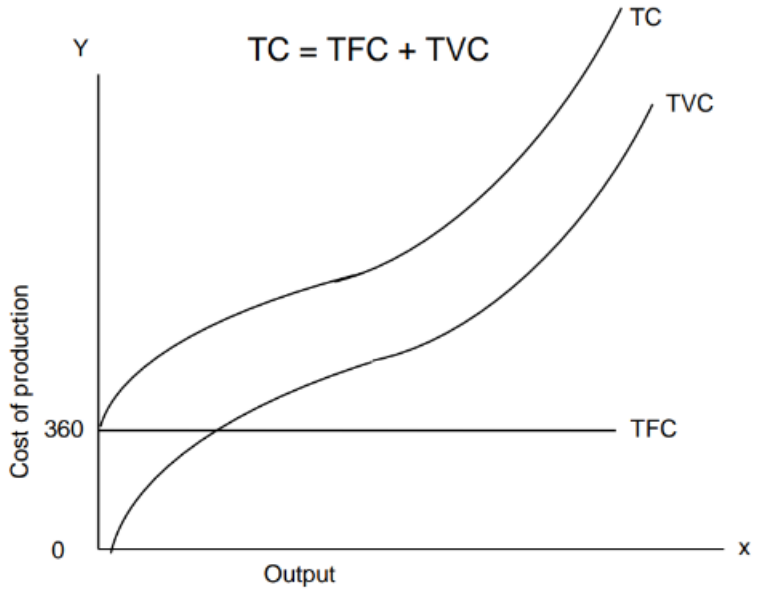


Figure 14: Total Cost Curve

The total cost curve rises upwards from left to right. In our example, the TC curve starts from Rs. 360 because even if there is no output, TFC is a positive amount. TC and TVC have the same shape because an increase in output increases them both by the same amount since TFC is constant. TC curve is derived by adding up vertically the TVC and TFC curves. The vertical distance between TVC curve and TC curve is equal to TFC and is constant throughout because TFC is constant.

Average fixed cost (AFC)

Average fixed cost is the fixed cost per unit of output. When TFC is divided by total units of output, AFC is obtained, Thus, AFC = TFC/Q.

AFC and output have an inverse relationship. It is higher at smaller levels and lower at higher levels of output in a given plant. The reason is simple to understand. Since AFC = TFC/Q, it is a pure mathematical result that the numerator remains unchanged, the increasing denominator causes diminishing cost. Hence, with the increase in output, TFC spreads over each unit of output. Consequently, AFC diminishes continuously. This relationship between output and fixed cost is universal for all types of business concerns.

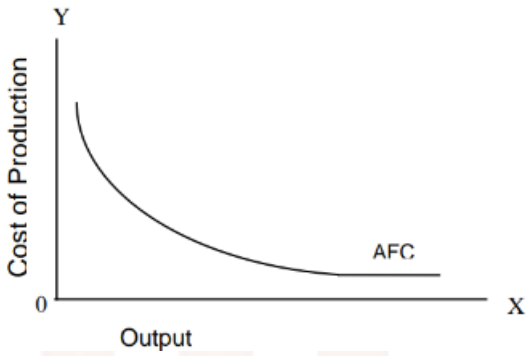


Figure 15: Average Fixed Cost Curve

The AFC curve has a negative slope. The curve slopes downwards throughout the length. The AFC curve goes very nearer to X axis, but never touches the X-axis. Graphically, it will fall steeply in the beginning, gently in middle and tend to become parallel to OX-axis. Mathematically speaking, as output increases, AFC diminishes. But AFC will never become zero because the TFC is a positive amount. AFC will never fall below a minimum amount because in the short run, plant capacity is fixed and output cannot be expanded to an unlimited extent.

Average variable cost (AVC)

The average variable cost is variable cost per unit of output. AVC can be computed by dividing the TVC by total units of output. Thus, AVC = TVC/Q. The AVC will come down in the beginning and then rise as more units of output are produced with a given plant. This is because, as we add more units of variable factors in a fixed plant, the efficiency of the inputs first increases and then decreases. Figure 5 depicts the average variable cost curve.

The AVC curve is a U-shaped cost curve. It has three phases.

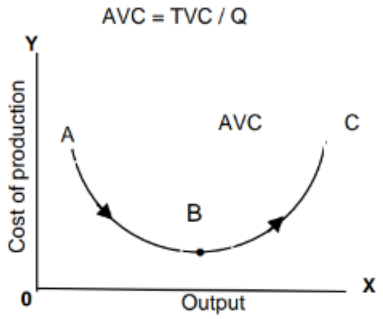


Figure 16: Average Variable Cost Curve

a) Decreasing phase – In the first phase from A to B, AVC declines. As the output expands, the AVC declines because when we add more quantity of variable factors to a given quantity of fixed factors, output increases more efficiently and more than proportionately due to the operation of increasing returns.

b) Constant phase – In the second phase, i.e., at B, AVC reaches its minimum point. When the proportion of both fixed and variable factors are ideal, the output will be optimal. After the firm operates at its normal full capacity, output reaches its zenith and as such AVC will reach its minimum point.

c) Increasing phase – In the third phase, from B to C, AVC rises. After the normal capacity is crossed, the AVC rises sharply. This is because additional units of variable factors will not result in more than proportionate output. Hence, greater output may be obtained but at much greater AVC. It is clear that as long as increasing returns operate, AVC falls and when diminishing returns set in, AVC tends to increase.

Average total cost (ATC) or average cost (AC)

AC refers to cost per unit of output. AC is also known as the unit cost since it is the cost per unit of output produced. AC is the sum of AFC and AVC. Average total cost or average cost is obtained by dividing the total cost by total output produced. AC = TC/Q. Also, AC is the sum of AFC and AVC.

In the short run, AC curve also tends to be U-shaped. The combined influence of AFC and AVC curves will shape the nature of AC curve.

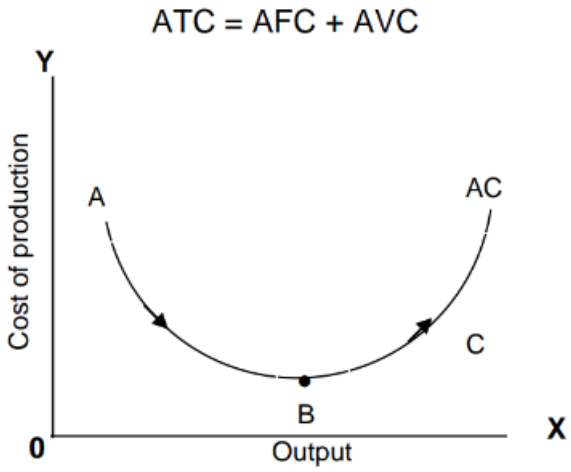


Figure 17: Average Cost Curve

As we observe, average fixed cost begins to fall with an increase in output while average variable cost declines and subsequently rises. As long as the falling effect of AFC is much more than the rising effect of AVC, the AC tends to fall. At this stage, increasing returns and economies of scale operate and complete utilization of resources forces the AC to fall.

When the firm produces the optimum output, AC drops to its minimum. This is called the least–cost output level. Again, at the point where the rise in AVC exactly counterbalances the fall in AFC, the balancing effect causes AC to remain constant.

In the third stage, when the rise in average variable cost is more than drop in AFC, then the AC shows a rise, when output is expanded beyond the optimum level of output, diminishing returns set in and diseconomies of scale starts operating. At this stage, the indivisible factors are used in sub- optimal proportions. Thus, AC starts rising in the third stage.

The short run AC curve is also called "plant curve". It indicates the optimum utilization of a given plant or optimum plant capacity.

Marginal cost (MC)

Marginal Cost may be defined as the net addition to the total cost as one more unit of output is produced. In other words, it implies additional cost incurred to produce an additional unit. For example, if it costs Rs. 100 to produce 50 units of a commodity and Rs. 105 to produce 51 units, then MC would be Rs. 5. It is obtained by calculating the change in total costs as a

result of a change in the total output. Also, MC is the rate at which total cost changes with output. Hence, MC = Δ TC / Δ TQ. Where, Δ TC stands for change in total cost and Δ TQ stands for change in total output.

Also, MCn = TCn –TC n-1

It is necessary to note that MC is independent of TFC and it is directly related to TVC as we calculate the cost of producing only one unit. In the short run, the MC curve also tends to be U-shaped.

The shape of the MC curve is determined by the laws of returns. If MC is falling, production will be under the conditions of increasing returns and if MC is rising, production will be subject to diminishing returns. Figure 7 depicts the marginal cost curves.

Figure 18: Marginal Cost Curve

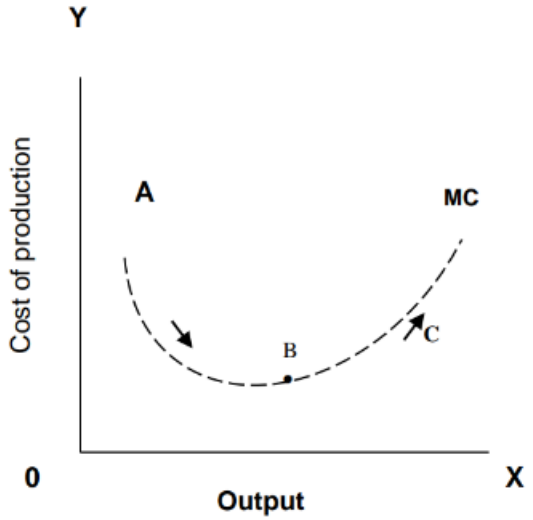


Table 7: Relationship between Average Cost and Marginal Cost

Output in Units	TC in Rs.	AC in Rs.	Difference MC (Rs.)
1	150	150	–
2	190	95	40
3	220	73.3	30
4	236	59	16
5	270	54	34
6	324	54	54
7	415	59.3	91
8	580	72.5	165

Relation between AC and MC

The **Average Cost (AC)** curve is typically **U-shaped** due to the behavior of **Fixed Costs** and **Variable Costs** as production increases.

Why is the AC curve U-shaped:

Decreasing Average Costs (initially): At the beginning of production, as output increases, the fixed costs are spread over more units, causing the Average Fixed Cost (AFC) to decline. Additionally, variable costs may initially increase at a decreasing rate due to increasing returns to scale or more efficient use of resources. Together, these forces lead to a reduction in Average Total Cost (AC) as output increases, which causes the left side of the AC curve to slope downward.

Increasing Average Costs (later): As production continues to increase, however, the law of diminishing returns comes into play. The marginal cost of production starts to increase, and variable costs (such as labor and materials) begin to rise at an increasing rate. This increase in variable costs eventually causes Average Variable Costs (AVC) to rise, and since AC = AVC + AFC, the AC curve starts to slope upwards after reaching its minimum point.

Thus, the U-shape of the AC curve is a result of both the initial decline in average costs due to economies of scale and the later rise in costs due to diminishing returns.

Relationship between AC, AFC, AVC, and MC:

1. **AC and AFC:**
    a. **AC = AFC + AVC**.
    b. **AFC (Average Fixed Cost)** decreases as output increases because fixed costs are spread over a larger number of units.
    c. As **AFC** decreases, it pulls the **AC** curve downwards initially, contributing to the downward-sloping portion of the U-shaped AC curve.
2. **AC and AVC:**
    a. **AVC** represents the **Average Variable Cost**, which includes costs that vary with output (e.g., raw materials and labor).
    b. **AC = AVC + AFC**, so **AC** will always be at least as large as **AVC**.
    c. The **AVC** curve also tends to be U-shaped because of diminishing returns: initially, AVC falls as production becomes more efficient, but after a certain point, AVC starts rising as the cost of additional units of output increases.

3. **AC and MC (Marginal Cost):**
    a. **MC** is the additional cost incurred when producing one more unit of output.
    b. When **MC < AC**, it pulls the AC down, causing the AC curve to slope downward. This happens when increasing production is relatively cheap, leading to economies of scale.
    c. When **MC > AC**, it pushes the AC up, causing the AC curve to slope upward. This occurs when increasing production becomes more costly due to diminishing returns.
    d. The **MC curve** intersects the **AC curve** at its minimum point. This is because when marginal cost is below average cost, average cost is falling, and when marginal cost is above average cost, average cost is rising.

The U-shape of the AC curve is due to the initial decreasing average costs (due to economies of scale) followed by increasing average costs (due to diminishing returns).

AC = AFC + AVC, so the decline in AFC initially helps reduce AC, while AVC rises due to diminishing returns.

The MC curve intersects the AC curve at its lowest point, as MC drives changes in AC. When MC is less than AC, AC decreases, and when MC is greater than AC, AC increases.

Figure 8 depicts the relationship between average cost and marginal cost.

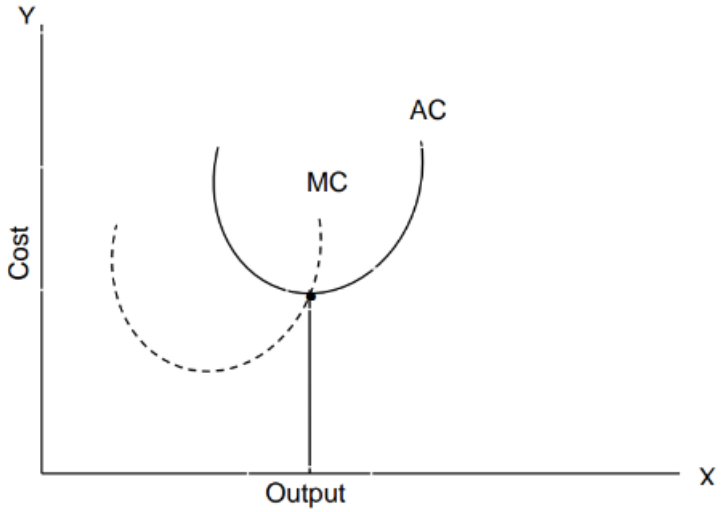


Figure 19: Relationship between Average cost and Marginal Cost

From the diagram, it is clear that:

1. Both MC and AC fall up to a certain range of output and rise afterwards.

2. When AC falls, MC also falls but at certain range of output, MC tends to rise even though AC continues to fall. However, MC would be less than AC. This is because MC is attributed to a single unit whereas in the case of AC, the decreasing AC is distributed over all the units of output produced.

3. As long as AC is falling, MC is less than AC. Hence, MC curve lies below AC curve. It indicates that the fall in MC is more than the fall in AC. MC reaches its minimum point before AC reaches its minimum.

4. When AC is rising, after the point of intersection, MC will be greater than AC. This is because in case of MC, the increasing MC is attributed to a single unit, whereas in case of AC, the increasing AC is distributed over all the output produced.

5. So long as the AC is rising, MC is greater than AC. Hence, MC curve lies to the left side of the AC curve. It indicates that the rise in MC is more than the rise in AC.

6. MC curve intersects the AC curve at the minimum point of the AC curve. This is because, when MC decreases, it pulls AC down and when MC increases, it pushes AC up. When AC is at its minimum, it is neither being pulled down nor being pushed up by the MC. Thus, when AC is minimum, MC = AC. The point of intersection indicates the least cost combination point or the optimum position of the firm. At output Q1, the firm is working at its “optimum capacity” with lowest AC. There is scope for “maximum capacity” with rising cost, beyond Q1.

Thus, we have discussed the cost-output relationship in the short run.

11. COST-OUTPUT RELATIONSHIP IN THE LONG RUN

In this section, we will discuss the cost output relationship in the long run. The long run is defined as a period of time where adjustments to changed conditions are complete. It is actually a period during which the quantities of all factors, variable as well as fixed factors, can be adjusted. Hence, there are no fixed costs in the long run. In the short run, a firm has to carry on its production within the existing plant capacity, but in the long run, it is not tied up to a particular plant capacity. If demand for the product increases, it can expand output by enlarging its plant capacity. It can construct new buildings or hire them, install new machines, employ administrative and other permanent staff. It can make use of the existing as well as new staff in the most efficient way and there is a lot of scope for transforming indivisible factors into divisible factors. On the other hand, if demand for the product declines, a firm can cut down its production permanently. The size of the plant can also be reduced and other expenditure can be minimized. Hence, production cost comes down to a greater extent in the long run.

As all costs are variable in the long run, the total of these costs is the total cost of production. Hence, the distinction between fixed and variables costs in the total cost of production will disappear in the long run. In the long run, only the average total cost is important and considered in taking long-term output decisions.

Long run average cost is the long run total cost divided by the level of output. In brief, it is the per-unit cost of production of different levels of output by changing the size of the plant or scale of production.

The long run cost–output relationship is explained by drawing a long run cost curve through short–run curves as the long period is made up of many short–periods just as the day is made up of 24 hours and a week is made out of 7 days. This curve explains how costs will change when the scale of production is varied. Figure 6.9 depicts a long run average cost curve.

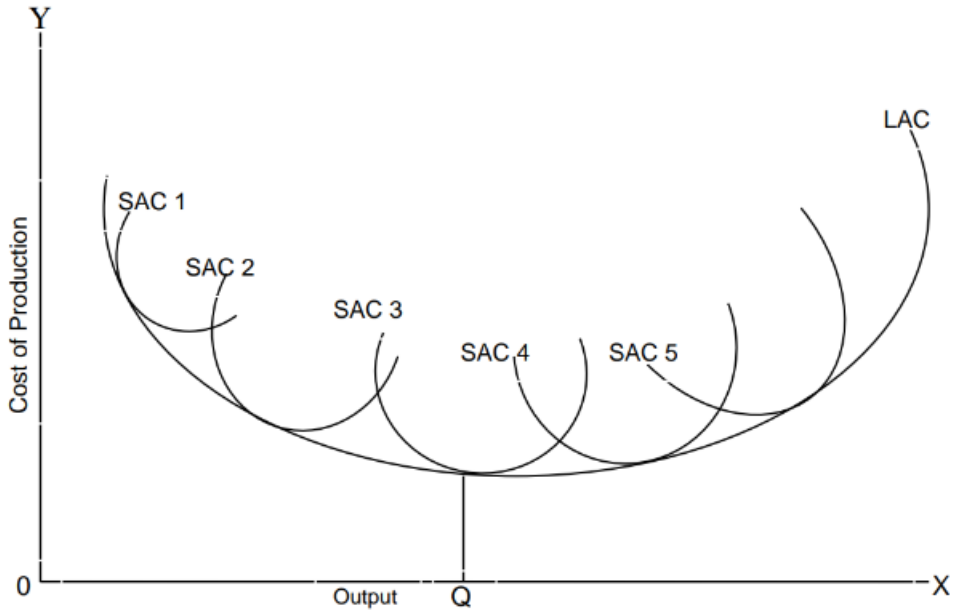


Figure 20: Changes in Cost with Variations on Scale of Production

The long run cost curves are influenced by the law of returns to scale as against the short run cost curves which are subject to the working of law of variable proportions.

In the short run, the firm is tied with a given plant and as such, the scale of operation remains constant. There will be only one AC curve to represent one fixed scale of output in the short run. In the long run, as it is possible to alter the scale of production, one can have as many AC curves as there are changes in the scale of operations.

In order to derive LAC curve, one has to draw a number of SAC curves, each curve representing a particular scale of output. The LAC curve will be tangential to the entire family of SAC curves. It means that it will touch each SAC curve at its minimum point.

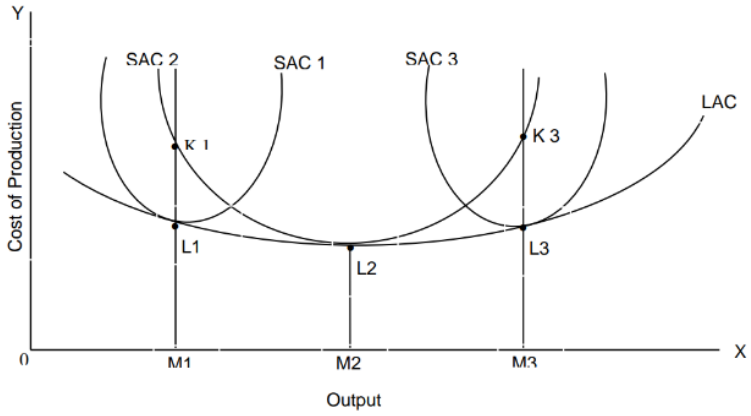


Figure 21: Production Cost Curves

Production cost in the short run and long run

Figure 10 depicts the production cost in the short run and in the long run.

In the diagram, the LAC curve is drawn on the basis of three possible plant sizes. Consequently, we have three different SAC curves – SAC1, SAC2 and SAC3. They represent three different scales of output. For output OM2, the AC will be L2M2 in the short run as well as the long run.

When output is to be expanded to OM3, it can be obtained at a higher average cost of production. K3M3 is the short run AC because scale of production would remain constant in the short run. But the same output of OM3 can be produced at a lower AC of L3M3 in the long run since the scale of production can be modified according to the requirements. The distance between K3L3 represents the difference between the cost of production in the short run and long run.

Similarly, when output is contracted to OM1 in the short run, K1M1 will become the short run AC and L1M1 will be the long run AC. Hence, K1L1 indicates the difference between short run and long run cost of production. If we join points L1, L2 and L3 we get LAC curve.

Important features of long run AC curves

The important features of long run AC curves are as follows:

1. Tangent curve – Different SAC curves represent different operational capacities of different plants in the short run. LAC curve is the locus of all these points of tangency. The SAC curve can never cut a LAC curve though they are tangential to each other. This implies that for any given level of output, no SAC curve can ever be below the LAC curve. Hence, SAC cannot be lower than the LAC in the long run. Thus, LAC curve is tangential to various SAC curves.

2. Envelope curve – It is known as envelope curve because it envelops a group of SAC curves appropriate to different levels of output.

3. Flatter U-shaped or dish-shaped curve – The LAC curve is also U shaped or dish-shaped cost curve. But it is less pronounced and much flatter in nature. LAC gradually falls and rises due to economies and diseconomies of scale.

4. Planning curve – The LAC curve is described as the planning curve of the firm because it represents the least cost of producing each possible level of output. This helps in producing an optimum level of output at the minimum LAC. This is possible when the entrepreneur is selecting the optimum scale plant. Optimum scale plant is that size where the minimum point of SAC is tangent to the minimum point of LAC.

5. Minimum point of LAC curve – It should be always lower than the minimum point of SAC curve. The LAC curve will touch the optimum plant SAC curve at its minimum point.

A rational entrepreneur would select the plant of optimum scale. Optimum scale plant is that size at which SAC is tangent to LAC, such that both the curves have the minimum point of tangency. In the diagram, OM2 is regarded as the optimum scale of output, as it has the least per unit cost. At OM2 output LAC = SAC.

LAC curves will be tangent to SAC curves lying to the left of the optimum scale or right side of the optimum scale. But at these points of tangency, neither LAC nor SAC is at its minimum. SAC curves are either rising or falling indicating a higher cost.

Managerial use of LAC

The study of LAC is of great importance in the managerial decision-making process. A few examples that quote the importance of studying LAC in the managerial decision-making process, are as follows:

1. It helps the management in the determination of the best size of the plant to be constructed or in getting the minimum cost output for a new plant.
2. The LAC curve helps a firm to decide the size of the plant to be adopted for producing the given output. For outputs less than cost lowering combination at the optimum scale i.e., when the firm is working subject to increasing returns to scale, it is more economical to use a slightly large plant operating at less than its minimum cost – output than to use a smaller unit. Conversely, at output beyond the optimum level, i.e., when the firm experiences decreasing returns to scale, it is more economical to use a slightly smaller plant than to use a slightly larger one. Thus, it explains why it is more economical to use a slightly small plant rather than to use a large plant.
3. LAC is used to show how a firm determines the optimum size of the plant. An optimum size of plant is one that helps in best utilization of resources in the most economical manner.

Thus, we have discussed features, importance and use of long run average costs.

Long run marginal cost

Figure 11 depicts the long run marginal cost curve.

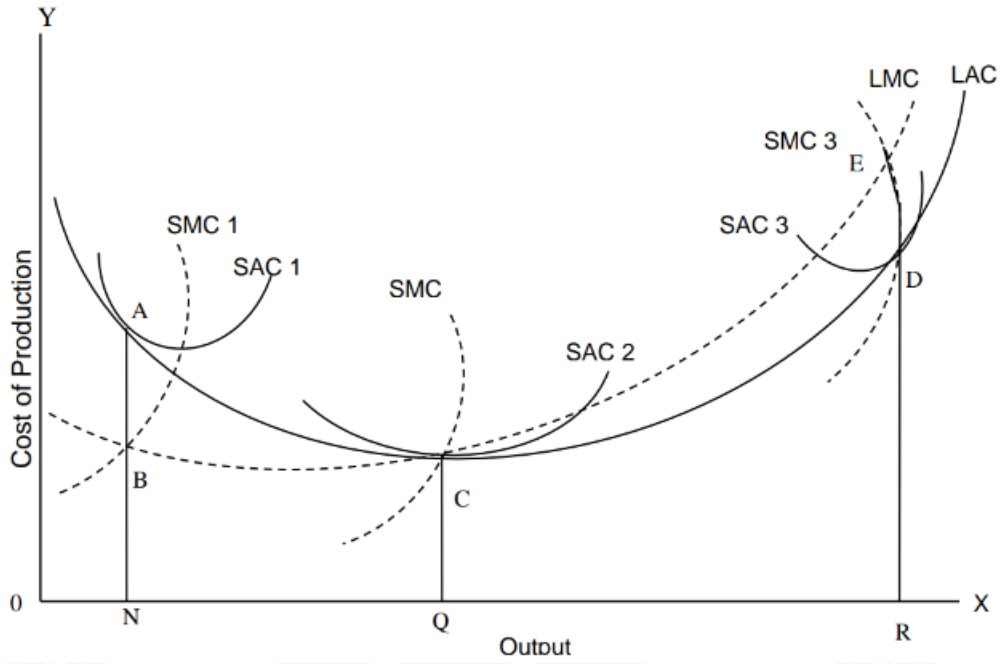


Figure 22: Long Run Marginal Cost Curve

A long-run marginal cost curve can be derived from the long-run average cost curve. Just as the SMC is related to the SAC, similarly the LMC is related to the LAC and, therefore, we can derive the LMC directly from the LAC. In the diagram we have taken three plant sizes and the corresponding three SAC and SMC curves. The LAC curve is drawn by enveloping the family of SAC curves. The points of tangency between the SAC and the LAC curves indicate different outputs for different plant sizes.

If the firm wants to produce ON output in the long run, it will have to choose the plant size corresponding to SAC1. The LAC curve is tangent to SAC1 at point A. For ON output, the average cost is NA and the corresponding marginal cost is NB. If LAC curve is tangent to SAC1 curve at point A, the corresponding LMC curve will have to be equal to SMC1 curve at point B. The LMC will pass through point B. In other words, where LAC is equal to SAC curve (for a given output) the LMC will have to be equal to a given SMC.

If output OQ is to be produced in the long run, it will be done at point C which is the point of tangency between SAC2 and the LAC. At point C, the short–run average cost (SAC2) and the short-run marginal cost (SMC2) are equal and, therefore, the LAC for output OQ is QC and the corresponding LMC is also QC. The LMC curve will, therefore, pass through point C.

Finally, for output OR, at point D the LAC is tangent to SAC3. For OR output at point E, LMC passes through SMC3. By connecting points B, C and E, we can draw the long-run marginal cost curve.

Costs of production: formulae

TC = Cost per unit x Total production. Or TC = TFC + TVC

TFC = (TC – TVC) or (AFC x Q)

TVC = (TC – TFC) or (AVC x Q) or addition of MC

AFC = (AC – AVC) or (TFC/Q)

AVC = (AC – AFC) or (TVC/Q) AC = (AFC + AVC) or (TC/Q)

MC = (TCn - TCn-1) or (Δ TC / Δ TQ)

SELF-ASSESSMENT QUESTIONS – 2

Fill in the blanks:	
14	Opportunity cost of anything is the alternative that has been _____.
15	Marginal cost deals with changes in cost of _____ unit whereas incremental cost deals with changes in cost of _____.
16	AC minus AVC would give us _____.
17	Total cost includes _____.
18	Marginal cost is associated with _____.
19	In the long run, all costs are _____.
True or False Questions:	
20	Average total cost increases if marginal cost is greater than average total cost. (True/False)
21	Marginal cost measures how total cost changes when input prices change. (True/False)
22	A fixed cost is a cost the firm must pay even if output is zero. (True/False)
23	A short-run cost function assumes that all inputs are fixed in supply (True/False)

12. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- In this unit, we have discussed the meaning of production, production function and its managerial uses. Production in economics implies transformation of inputs into outputs for our final consumption.
- Production function explains the quantitative relationship between the amounts of inputs used to get a particular physical quantity of outputs. The ratios between the two quantities are of great importance to a producer in taking his decisions in the production process.
- There are two kinds of production functions - short run and long run. In the case of short run production function, we come across a change in either one or two variable factor inputs while all other inputs are kept constant.
- The law of variable proportion explains how there will be variations in the quantity of output when there is change in only one variable factor input while all other inputs are kept constant. On the other hand, iso-quants and iso-cost curves explain how there will be changes in output when only two variable inputs are changed while all other inputs are kept constant. Under long run production function, the laws of returns to scale explain changes in output when all inputs, both variable as well as fixed changes in the same proportion.
- Economies of scale give information about the various benefits that a firm will get when it goes for large scale production. Economies of scope on the other hand tells us how there will be certain specific advantages when one firm produces more than two products jointly than two or three firms produce them separately. Diseconomies of scale and diseconomies of scope tell us that there are certain limitations to expansion in output.
- Cost analysis indicates the various amounts of costs incurred to produce a particular quantity of output in monetary terms.
- The various kinds of cost concepts help a manager to take the right decisions. The cost function explains the relationship between the amounts of costs to be incurred to produce a particular quantity of output.
- The short run cost function gives information about the nature and behavior of various cost curves. The long-run cost function tells us how it is possible to obtain more output at lower costs in the long run.

- Thus, the knowledge of both production function and cost functions helps a business executive to work out the best possible factor combinations to maximize output with minimum costs.

13. GLOSSARY

Economies of scale	-	Advantages or benefits that accrue to a firm as a result of increase in its scale of production.
Economies of scope	-	Benefits which arise to a firm when it produces more than one product jointly rather than producing two items separately by two different business units.
Iso cost line	-	Indicates the different combinations of the two inputs which the firm can purchase at given prices with a given outlay.
Isoquant/Iso product curve	-	A curve which shows the different combinations of the two inputs producing the same level of output.
Law of variable proportions	-	As the quantity of only one factor input is increased to a given quantity of fixed factors, beyond a particular point, the marginal, average and total output eventually decline.
Long run	-	Period of time wherein the producer will have adequate time to make changes in the factor combinations.
Marginal rate of technical substitution	-	Rate at which a factor of production can be substituted for another at the margin without affecting any change in the quantity of output.
Production	-	Transformation of physical Inputs into physical outputs.
Production function	-	Technological or engineering relationship between physical quantity of inputs employed and physical quantity of outputs obtained by a firm.
Short run	-	Period of time in which only the variable factors can be varied while fixed factors like plants, machinery, top management, etc. would remain constant.
Average cost	-	Cost per unit of output.

Average fixed cost	-	Fixed cost per unit of output.
Average variable cost	-	Variable cost per unit of output.
Cost function	-	Technical relationship between the cost and output.
Fixed costs	-	These are costs that do not vary with either expansion or contraction in output.
Marginal cost	-	Additional cost incurred to produce an additional unit of output. Net addition to the total cost as one more unit of output is produced.
Opportunity cost	-	Revenue which could have been earned by employing that good or service for some other alternative uses.
Total cost	-	The aggregate expenditure incurred by a firm to produce a given quantity of output.
Total fixed cost	-	Total money expenses incurred on fixed inputs like plant, machinery, tools and equipment in the short run.
Total variable cost	-	Total money expenses incurred on the variable factor inputs like raw materials, power, fuel, water, transport and communication, etc., in the short run.
Variable costs	-	Costs which directly and proportionately increase or decrease with the level of output produced.

14. TERMINAL QUESTIONS

1. Define production function and distinguish between short run and long run production function.
2. Discuss the uses of production function.
3. Explain the law of variable proportions.
4. Explain how a product would reach equilibrium position with the help of –iso-quants and iso-cost curve.
5. Discuss any one law of returns to scale with example.
6. Give a brief description of the following:
   A. Implicit and explicit cost
   B. Actual and opportunity cost
   C. Discuss the various determinants of costs.
   D. Explain cost output relationship with reference to:
   E. Total fixed cost and output
   F. Total variable cost and output
   G. Explain features of LAC curve with a diagram.

15. ANSWERS

Self-Assessment Questions

1. New utilities, addition
2. Technological, engineering
3. Variable
4. Highest
5. Various alternatives, particular
6. Diminishing returns
7. Firms' industry
8. One product jointly
9. False
10. True
11. False
12. True
13. True
14. Foregone
15. One, a group of units
16. AFC
17. Normal
18. Variable
19. Variable
20. True
21. False
22. True
23. False

Terminal Questions Answers

Answer 1: The entire theory of production centers revolves around the concept of production function. In short run Quantities of all inputs both fixed and variable will be kept constant and only one variable input will be varied, for example, law of variable proportions. In the long run the producer will vary the quantities of all factor inputs, both fixed as well as variable in the same proportion Refer to section 2.

Answer 2: It is of immense utility to the managers and executives in the decision- making process at the firm level Refer to section 2.

Answer 3: The law of variable proportions gives a key insight in determination of the ideal combination of factor inputs. All factor inputs are not available in plenty Refer to section 3.

Answer 4: Isoquant is also called equal product curve or product indifference curve or constant product curve. The Isoquant curve represents only one particular level of output. Refer to section 4.

Answer 5: The change in output when all factor inputs are changed or made available in the required quantity. Refer to section 5.

Answer 6: A. Explicit costs are those costs which are in the nature of contractual payments and are paid by an entrepreneur to the factors of production. Explicit cost is the earnings of owner-employed resources

Actual costs are those costs that involve financial expenditures at some time and hence, are recorded in the books of accounts. Opportunity cost of anything is the cost of displaced alternatives or costs of sacrificed alternatives.

Answer 7: The following factors deserve our attention in this connection: technology, rate of output, size of Plant and scale of production, prices of factor inputs.

A. Fixed costs in the short run remain constant because the firm does not change the size of the plant and the amount of fixed factors employed.

B. The costs corresponding to variable factors are discussed as variable costs.

Answer 8: The important features of long run AC curves are as follows: tangent curve, envelope curve, flatter U-shaped or dish-shaped curve, planning curve

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